AoE2 Civ Explorer

Code documentation

What each part of the code does, and why

30 October 2026 · 21 chapters · 356 functions and 1 object

This document explains the code of AoE2 Civ Explorer. Each file is a chapter, and the chapter names the folder the file is in. In the chapters on scripts, each function defined at the top level of its script is a subchapter: it gives the lines the function takes up, says what it does and why, and lists the functions that call it and the functions it calls. A function handed on to be called later, such as an event handler, counts as called by the function that hands it on. Helpers defined inside a function are part of that function's text.

"Exported" marks a function that other scripts import, and "internal" one that only its own script uses. The chapters on the page, its style sheets and the data are split into parts in the same way, each with its lines. Rules and requirements are those of RULES.md and REQUIREMENTS.md.

Every name of a file or a function links to its chapter or subchapter, and the page header always names the chapter. The UML diagram after the contents shows how the scripts use each other, and the index at the end lists every chapter and function in alphabetical order.

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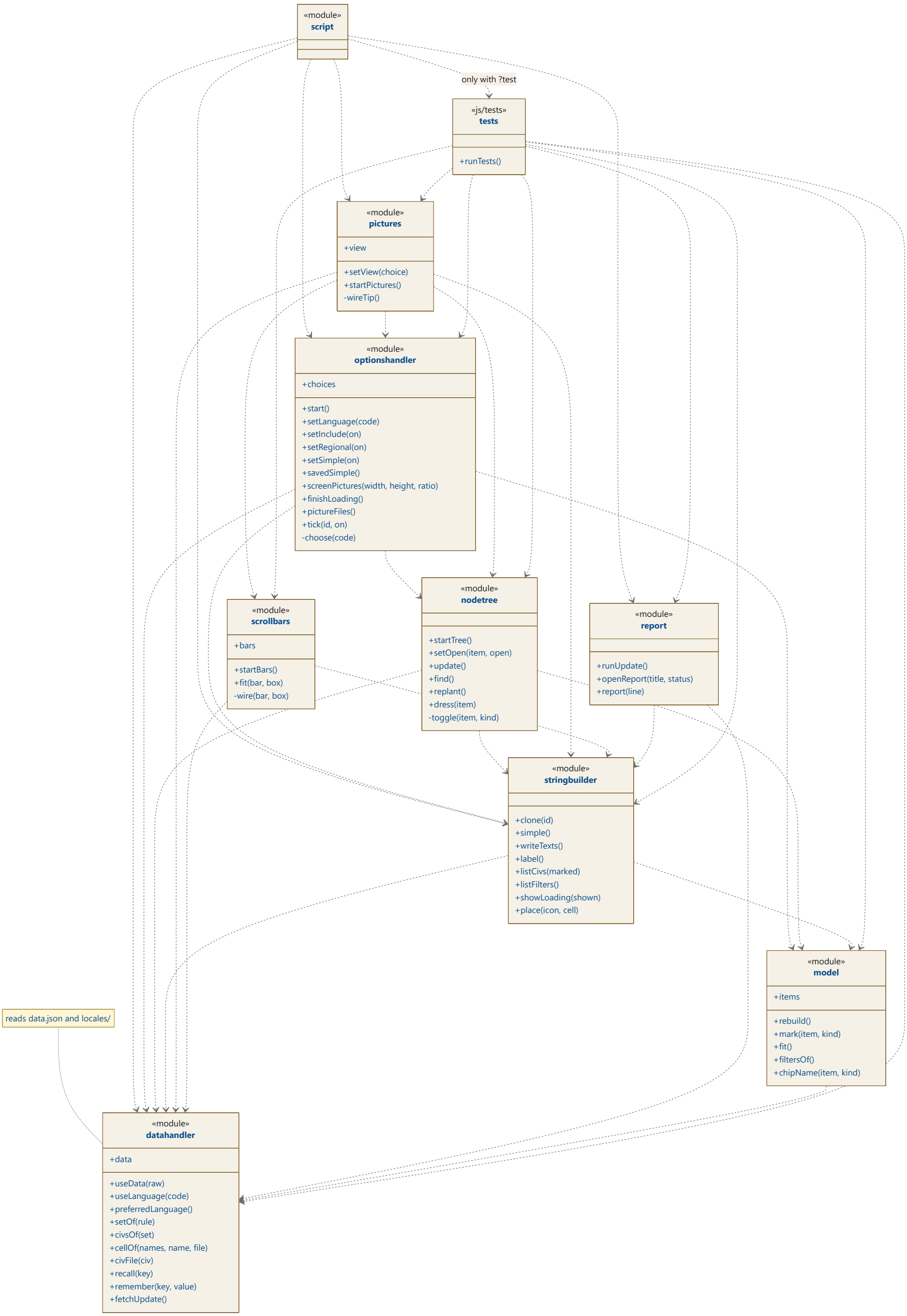
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UML diagram

Each arrow points from a script to a script it uses. + marks what a script shares with others, - what it keeps to itself. Each box links to the chapter of its script, and each name in it to its subchapter; the box *tests* stands for the scripts in `js/tests/` and links to js/tests/run_test.js.



1 index.html

Directory: the main folder 98 lines 11 parts
Loads: css/base.css, css/pictures.css and js/script.js

The page's structure, and all of its HTML (Rule 2). It holds no text of its own: the scripts fill in every word in the chosen language, and copy its templates instead of writing HTML.

1.1 Head

lines 1–13

Sets the language, character set, viewport, light and dark colour schemes, title and description, and links base.css, pictures.css and script.js, as a module. The tab icon starts as the empty `data:`, so no picture loads before the scripts run (Req. 33).

1.2 Loading screen

line 15

The screen shown without Simple UI until every picture has loaded, with its text. `showLoading()` adds the turning picture from its template.

1.3 Header

lines 16–19

Holds the title twice: the full sentence of Req. 2 for Simple UI, and "Civ Explorer" for the pictures (Req. 0). Then the cogwheel Options button (Req. 6), a text character with a hidden label. Hidden until the page starts.

1.4 Messages

lines 20–21

A noscript line for browsers without JavaScript, and a load error shown if data.json cannot be loaded, so a visitor is never left with a blank page.

1.5 Tree pane

lines 22–43

The start of the content, then the left pane, in the order of Req. 3: heading, Clear, Expand all and Collapse all, the Find field with its label, clearing cross and live match count, the must have and must lack legend, and the empty tree.

1.6 Civilisations pane

lines 44–57

The right pane (Req. 4): heading, live count, the Filters list (Req. 22), the civilisation list and the hint shown when it is empty. It scrolls on its own, so the page never does (Rule 12).

1.7 Footer

lines 58–59

Names and links the data source and its date (Req. 5), then Microsoft's notice, in English, with links to its rules and to the project. The code writes the sentence around the links.

1.8 Options window

lines 60–76

A dialog (Req. 7, 8, 32): title, the language label, button and list, then the settings: Simple UI, Include Unique and Universal with its description, Include Regionals, Civ view and Tooltips; and Cancel and OK.

1.9 Report window

lines 77–85

A dialog for the reports of the tests and the updater: title, live status, result list, download links and Close. Always in English, so it is marked `lang="en"`.

1.10 Templates

lines 86–95

Markup for tree rows, Filters buttons, civilisations, the loading picture, the icons of civilisations and of the tree, language options, report lines, downloads and the scroll bar. The code copies these instead of writing HTML (Rule 2).

1.11 Tooltip

line 96

The note that follows the mouse without Simple UI, over the tree, Include or a civilisation. Hidden otherwise.

2 base.css

Directory: css/ 201 lines 12 parts
Loaded by: index.html

How the page looks with Simple UI, and its layout either way: colours, sizes and layouts for big and small screens, in light and dark mode. Every rule with a picture or a font is in pictures.css instead.

2.1 Colours

lines 1–33

Named colours for light mode, redefined for dark mode: text, lines, marks, hover, Find's yellow, the green and red of the marks, and the dark box behind the white dash; and the 44-pixel target size and the font. The tests measure their contrast.

2.2 Base

lines 34–46

Box sizing, `hidden` that always hides, a full-height page that never scrolls (Rule 12), the system font with 1.5 line spacing (WCAG 1.4.8), a 3 px focus outline (WCAG 2.4.13), and a class that hides text from sight but not from screen readers.

2.3 Header and language

lines 48–66

Lays out the title and cogwheel, and in the Options window the language label, its button with a turning chevron, and the list of languages, where the current one is marked with a bar and bold.

2.4 Panes

lines 68–83

Two panes side by side from 720 px wide; below that they share a phone's height, and the footer is hidden. Each pane's head stays in place while its list scrolls.

2.5 Controls

lines 84–101

Buttons and the Find field are at least 44 by 44 px (WCAG 2.5.5), with visible borders and a hover colour. Find's message keeps its column, so the box never changes size while typing (Rule 18). The legend, and the scrolling lists.

2.6 Tree

lines 103–121

Indented rows with dotted guide lines, 44 px rows, turning arrows, pressed and greyed-out rows, and found rows in yellow with the light colours (Req. 20) whatever the colour scheme.

2.7 Check boxes

lines 122–140

Marks drawn with borders (Req. 14, 23, 26): a green tick, a red cross, and a white dash on a dark box, in system colours when colours are forced.

2.8 Names and views

lines 141–145

Hides Civ view and Tooltips, which only the pictures use; makes top-level and group names bold; keeps counts on one line at the right; and draws focus rings inside rows.

2.9 Filters and list

lines 147–163

Filters as a wrapping row of rounded chips under the pane's head, the civilisation list in columns that never split a name, the empty hint and the footer.

2.10 Dialogs

lines 165–185

Size, padding, border and backdrop of the Options and report windows. The Options window is a grid of fixed height, so opening the language list never resizes it (Rule 18); only the report's list scrolls.

2.11 Small screens

lines 187–198

On short screens everything scrolls as one, as at 400% zoom (WCAG 1.4.10); on narrow ones the padding and the tree's indent shrink.

2.12 Test helper

lines 200–201

The wider text spacing the tests apply (WCAG 1.4.12). Used only while testing.

3 pictures.css

Directory: css/ 225 lines 14 parts
Loaded by: index.html

How the page looks without Simple UI. Every rule here that shows a picture starts with `.pictures` or `html.pictures`, so Simple UI turns them all off (Rules 9, 22), and picture addresses go from this folder, as `../assets/`. The `[data-assets="4k"]` and `[data-background]` rules pick the pictures for the screen.

3.1 **Parchment and cursors** lines 1–24
The parchment behind the page, fitted so the content lies inside its frame on any window, in the four sizes; the sword and the pointing hand as cursors, the `_x2` ones drawn at half their size; and how far the tooltip sits from the pointer.

3.2 **Header and footer** lines 25–47
The footer's colours, the title in its frame with the short title, the load error, and the Options button from its three-state sheet.

3.3 **Fonts** lines 48–57
Georgia, Times New Roman Bold and Trajan Pro Bold from `assets/fonts/`, under their own names, trying the installed copy first. Only rules for the pictures use them, so Simple UI never downloads a font.

3.4 **Text buttons** lines 58–67
Every button with words, but those of the report, shows the button picture, normal, under the mouse or pressed, with its ends drawn apart so it can be any width.

3.5 **Loading screen** lines 69–72
The screen that covers the page while the pictures load, and its turning picture, the only animation, which stops with reduced motion (Rule 11).

3.6 **Boxes and radio buttons** lines 74–90
Check boxes from their sheet, a row for each state and a column for normal, under the mouse and pressed, with a black dash; Tooltips and the Civ view radio buttons, shown only here.

3.7 **Frames** lines 91–98
The 9-slice frames of the two panes, and their fonts and colours.

3.8 **Options window** lines 99–127
The window's frame and the parchment behind it, measured to fit around the same window as with Simple UI, with OK and Cancel in the parchment's corner and the title on its ribbon on tall enough windows.

3.9 **Tree** lines 128–149
The tree's colours, found rows in amber, the rows' icons from the four atlases with an outline by kind and reach as `colour_key.csv` says, and greyed-out icons.

3.10 **Tools** lines 150–158
Filters chips, no legend, Find first, and Clear and Expand all as two buttons that share the pane's width.

3.11 **Find box** lines 159–176
Find's field in its 9-slice frame, 325 px wide, with the search picture while empty and the clearing cross while not; the hidden label and Include's hidden description.

3.12 **Tooltip** lines 177–181
The tooltip's place, set by the script as numbers, and its parchment look.

3.13 **Civilisations** lines 182–197
The civilisations' icons from their atlas, and the tiles, six to a row, each with its icon and its name in Trajan Pro.

3.14 Scroll bars

lines 198–225

The game's scroll bars: the browser's are hidden, room is kept for the bar, and the bar is placed, sized and drawn from its arrows, body and knob, each from its sheet, as `fit()` sets them.

4 script.js

Directory: js/ 24 lines no functions
Imports: js/model/datahandler.js, js/view/stringbuilder.js, js/view/pictures.js, js/view/report.js, js/controller/optionshandler.js, and js/tests/run_test.js (only with `?test`)
Imported by: index.html

Where the page starts. It defines no functions; its top-level code runs once.

4.1 `Start-up` TOP-LEVEL CODE

lines 7–24

Sets Simple UI from the saved choice first, so pictures load at once or never. Loads data.json and the visitor's language (all of them for `?test` or `?update`), then starts the page and its pictures, and lets the loading screen go; `?test` loads the tests and `?update` runs the updater. If loading fails, it shows the load error.

Called by *no function of the code*
Calls in js/model/datahandler.js: `load()`, `useData()`, `loadLanguage()`, `loadLanguages()`, `preferredLanguage()`; in js/view/stringbuilder.js: `showLoadError()`, `showLoading()`; in js/view/pictures.js: `startPictures()`; in js/view/report.js: `runUpdate()`; in js/controller/optionshandler.js: `savedSimple()`, `finishLoading()`, `setSimple()`, `start()`; in js/tests/run_test.js: `loadFailed()`, `runTests()`

5 datahandler.js

Directory: js/model/ 345 lines 46 functions, 31 exported
Imports: nothing
Imported by: js/script.js, js/model/model.js, js/view/stringbuilder.js, js/view/pictures.js, js/view/scrollbars.js, js/view/report.js, js/controller/nodetree.js, js/controller/optionshandler.js, js/tests/run_test.js, js/tests/helpers_test.js, js/tests/static_test.js, js/tests/behaviour_test.js, js/tests/interface_test.js, js/tests/pictures_test.js, js/tests/audit_test.js

Everything about data: loading, the chosen language and its text, sets of civilisations, saved choices, and the updater.

5.1 `attempt(action)` EXPORTED

lines 9–15

Runs a function and returns null instead of an error. Used where the browser may refuse, such as storage in a private window, so a refusal never breaks the page.

Called by `recall()`, `remember()`; in js/view/scrollbars.js: `hold()`, `drag()`; in js/controller/optionshandler.js: `rooted()`, `pictureFiles()`, `choose()`; in js/tests/run_test.js: `restore()`; in js/tests/interface_test.js: `optionTests()`; in js/tests/pictures_test.js: `tipLooks()`, `simpleSwitch()`
Calls *no function of the code*

5.2 `recall(key)` EXPORTED

line 17

Reads a saved choice from the browser's storage, or null. Remembers the visitor's language and options between visits.

Called by `preferredLanguage()`; in js/view/pictures.js: `startPictures()`; in js/controller/optionshandler.js: `savedSimple()`, `start()`; in js/tests/run_test.js: `prepare()`; in js/tests/interface_test.js: `englishFirst()`, `optionTests()`; in js/tests/pictures_test.js: `tipLooks()`, `simpleSwitch()`
Calls `attempt()`

5.3 `remember(key, value)` EXPORTED

lines 19–21

Saves a choice in the browser's storage, or removes it when the value is null. Failures are ignored, as the page works without storage.

Called by in js/view/pictures.js: `setView()`, `setTips()`; in js/controller/optionshandler.js: `setInclude()`, `setRegional()`, `setSimple()`, `choose()`; in js/tests/run_test.js: `restore()`; in js/tests/interface_test.js: `englishFirst()`
Calls `attempt()`

5.4 `load(url)` EXPORTED

lines 23–31

Fetches a file as text and returns a promise. It uses XMLHttpRequest because, unlike `fetch()`, it also reads files when the page is opened from disk, as the tests in CI do.

Called by `loadLanguage()`, `fetchUpdate()`; in js/script.js: `Start-up`; in js/tests/run_test.js: `rounds()`; in js/tests/helpers_test.js: `edges()`; in js/tests/static_test.js: `scripts()`, `styles()`
Calls *no function of the code*

5.5 `fill(template, values)` EXPORTED

lines 33–34

Replaces `{name}` placeholders in a text with values, leaving unknown ones alone. All text lives in the data with placeholders, so the code never builds sentences itself.

Called by `plural()`, `refresh()`; in js/model/model.js: `chipName()`; in js/view/stringbuilder.js: `listFilters()`; in js/view/report.js: `runUpdate()`; in js/controller/nodetree.js: `count()`; in js/tests/run_test.js: `reportLine()`, `runTests()`; in js/tests/behaviour_test.js: `fullName()`, `marking()`, `counts()`; in js/tests/interface_test.js: `languageTests()`
Calls *no function of the code*

5.6 `treeKeys(nodes, key)` EXPORTED

lines 36–38

Lists every item (or group) name in a tree, top to bottom. Used to check the tree, to find groups that need text, and by the updater to know which items to look up.

Called by `missingText()`, `refresh()`; in js/tests/helpers_test.js: `keys()`
Calls *no function of the code*

5.7 `useLanguage(code)` EXPORTED lines 47–54

Switches the active language: its text, language tag, sorting, plural rules and number format. Everything that shows, sorts or counts text reads these, so one call changes them all.

Called by in js/controller/optionshandler.js: `setLanguage()`
Calls *no function of the code*

5.8 `nameOf(item)` EXPORTED line 56

Returns a tree item's or group's name in the current language, or its key if missing. Used wherever an item is shown or searched.

Called by in js/model/model.js: `share()`, `chipName()`; in js/view/stringbuilder.js: `label()`; in js/controller/nodetree.js: `find()`; in js/tests/helpers_test.js: `fixtures()`; in js/tests/behaviour_test.js: `fullName()`, `branches()`, `lineTops()`, `standIns()`, `counts()`, `tree()`, `finding()`; in js/tests/interface_test.js: `languageTests()`, `findKey()`, `uniqueFilterTests()`; in js/tests/pictures_test.js: `findLooks()`; in js/tests/audit_test.js: `controlChecks()`
Calls *no function of the code*

5.9 `civName(civ)` EXPORTED line 58

Returns a civilisation's name in the current language, or its key if missing. Keeps the civilisation list translated.

Called by in js/view/stringbuilder.js: `listCivs()`; in js/tests/helpers_test.js: `expected()`
Calls *no function of the code*

5.10 `iconFile({ kind, picture })` INTERNAL line 60

Returns the path of a tree item's icon PNG from its kind and picture number, such as `assets/images/icons/nodes/unit/8.png`, for the updater's report of new icons.

Called by `addIcons()`
Calls *no function of the code*

5.11 `civFile(civ)` EXPORTED line 62

Returns the name of a civilisation's icon file: its name in lower case, unless `icons.names` in data.json gives another, such as `inca` for the Incas.

Called by in js/view/stringbuilder.js: `civIcon()`; in js/tests/pictures_test.js: `unpictured()`
Calls *no function of the code*

5.12 `byFile(a, b)` INTERNAL line 64

Sorts picture numbers or names as their PNG files sort by name, `1.png`, `10.png`, `100.png` and on to `11.png`, which is the order of the cells in an atlas.

Called by `cellOf()`, `addIcons()`
Calls *no function of the code*

5.13 `cellOf(names, name, file = String)` EXPORTED lines 68–73

Finds a picture's cell in its atlas, which holds the pictures in file-name order, row by row, with as many columns as the square root of their number, rounded up. The order is worked out once per list.

Called by in js/view/stringbuilder.js: `civIcon()`; in js/controller/nodetree.js: `dress()`
Calls `byFile()`

5.14 `plural(key, n)` EXPORTED lines 75–78

Picks the right form of a counted text, such as "1 civilisation" or "5 civilisations", by the language's plural rules, and fills in the formatted number.

Called by in js/view/stringbuilder.js: `listCivs()`; in js/controller/nodetree.js: `count()`, `find()`; in js/tests/behaviour_test.js: `marking()`, `counts()`, `finding()`
Calls `fill()`

5.15 `plain(text)` EXPORTED lines 80–81

Lower-cases a text and strips accents, spaces and punctuation, so Find matches names however they are typed, in any language.

Called by in js/controller/nodetree.js: `find()`; in js/tests/behaviour_test.js: `finding()`; in js/tests/interface_test.js: `findKey()`
Calls *no function of the code*

5.16 `localeFile(code)` EXPORTED line 83

Returns the path of a language's file, such as `locales/de.json`. Keeps the pattern in one place for loading and for the updater's downloads.

Called by `loadLanguage()`, `pack()`; in js/tests/static_test.js: `updaterTests()`
Calls *no function of the code*

5.17 `unpack(file)` EXPORTED lines 85–88

Turns data.json as stored into the shape used in memory, where overrides and text are kept per language, so other languages can be added beside English.

Called by `useData()`; in js/tests/static_test.js: `updaterTests()`
Calls *no function of the code*

5.18 `addLanguage(set, code, file)` EXPORTED lines 90–93

Adds one language file's overrides and text to the data in memory. Used when a language loads, and when a test reads back the files the updater would write.

Called by `loadLanguage()`; in js/tests/static_test.js: `updaterTests()`
Calls *no function of the code*

5.19 `useData(raw)` EXPORTED lines 95–99

Reads data.json and gives each civilisation one bit, so a set of civilisations is a single number that `&` and `|` combine quickly.

Called by in js/script.js: `Start-up`
Calls `unpack()`

5.20 `loadLanguage(code)` EXPORTED lines 101–103

Loads a language's file unless it is loaded already. A visitor downloads only the language shown, not all of them.

Called by `loadLanguages()`; in js/script.js: `Start-up`; in js/controller/optionshandler.js: `choose()`
Calls `load()`, `localeFile()`, `addLanguage()`

5.21 `loadLanguages()` EXPORTED line 105

Loads every language's file. The tests and the updater need them all.

Called by in js/script.js: `Start-up`
Calls `loadLanguage()`

5.22 `pack(set)` EXPORTED lines 107–118

Splits the data in memory back into files: data.json with English, and one file per other language. The updater writes these, and a test checks that nothing is lost.

Called by `fetchUpdate()`; in js/tests/static_test.js: `updaterTests()`
Calls `localeFile()`

5.23 `preferredLanguage()` EXPORTED lines 120–123

Chooses the language to start in: the one in the address (`?lng=`), else the visitor's saved choice, else English (Req. 10). Shared links and return visits keep their language; first visits are English.

Called by in js/script.js: `Start-up`; in js/controller/optionshandler.js: `start()`; in js/tests/interface_test.js: `englishFirst()`
Calls `recall()`

5.24 `bit(civ)` INTERNAL lines 125–128

Returns a civilisation's bit, or stops with an error for an unknown name, so a typo in data.json cannot silently drop a civilisation.

Called by `setOf()`
Calls *no function of the code*

5.25 `size(set)` EXPORTED lines 130–134

Counts the civilisations in a set by counting its bits. Behind every number on the page.

Called by in js/model/model.js: `prune()`; in js/controller/nodetree.js: `count()`; in js/tests/helpers_test.js: `fixtures()`; in js/tests/behaviour_test.js: `marking()`, `crowding()`, `beyond()`, `counts()`; in js/tests/interface_test.js: `optionTests()`; in js/tests/pictures_test.js: `simpleSwitch()`
Calls *no function of the code*

5.26 `setOf(rule)` EXPORTED lines 136–140

Turns an availability rule from data.json (`"all"`, a `has` list or a `lacks` list) into a set. Rules stay short in the file while the code works with sets.

Called by in js/model/model.js: `collect()`, `prune()`; in js/tests/static_test.js: `updaterTests()`
Calls `bit()`

5.27 `civsOf(set)` EXPORTED line 142

Turns a set back into a list of civilisations, in data.json order. Used to list the civilisations that fit.

Called by in js/view/stringbuilder.js: `listCivs()`; in js/tests/helpers_test.js: `expected()`; in js/tests/static_test.js: `updaterTests()`
Calls *no function of the code*

5.28 `isScalar(value)` INTERNAL line 144

Tells whether a value is text, a number, true or false, or null. Helps `toJson()` decide what fits on one line.

Called by `isFlatList()`, `flat()`, `toJson()`
Calls *no function of the code*

5.29 `isFlatList(value)` INTERNAL line 146

Tells whether a value is a list of scalars only, which `toJson()` writes on one line.

Called by `toJson()`
Calls `isScalar()`

5.30 `flat(value)` INTERNAL lines 148–152

Writes a value as one line of JSON, with a space after each comma and colon. The building block of `toJson()`.

Called by `toJson()`
Calls `isScalar()`

5.31 `toJson(value, indent = "")` EXPORTED lines 154–171

Writes JSON the way data.json is laid out: short objects and lists on one line, the rest indented, and a list that ends an object of scalars opened on the object's line. Keeps the updater's files readable and their changes easy to review.

Called by `fetchUpdate()`; in js/tests/static_test.js: `updaterTests()`
Calls `isScalar()`, `isFlatList()`, `flat()`

5.32 `isLower(char)` INTERNAL line 173

Tells whether a character is a lower-case letter. `joinLines()` uses it to tell a hyphenated word from a real hyphen.

Called by `joinLines()`
Calls *no function of the code*

5.33 `capitalise(text)` INTERNAL line 175

Upper-cases the first letter of a text, as some names in the game start in lower case.

Called by `fromExtra()`, `localName()`, `translate()`
Calls *no function of the code*

5.34 `fromExtra(name, extra)` EXPORTED lines 177–181

Turns an entry of `extras` in data.json, an item aoe2techtree.net gives to no civilisation, into the node, owners and picture counts the updater gathers for every other item. A test checks the extras the same way.

Called by `addExtras()`; in js/tests/static_test.js: `badExtras()`
Calls `capitalise()`

5.35 `joinLines(text, settings)` EXPORTED lines 183–194

Joins a game text broken with
 into one line, by the language's rules for spaces and hyphens, and tidies spaces. The game breaks names to fit its buttons.

Called by `localName()`, `translate()`; in js/tests/static_test.js: `updaterTests()`
Calls `isLower()`

5.36 `localName(strings, node, settings, offset)` INTERNAL lines 196–201

Finds an item's name in a language's game texts. If that name looks abbreviated, it takes the bold name from the item's help text instead, which is written out in full.

Called by `translate()`
Calls `joinLines()`, `capitalise()`

5.37 `availabilityOf(owners, civs)` EXPORTED lines 203–208

Writes who has an item in its shortest form: `"all"`, a `has` list or a `lacks` list. Keeps data.json small; a test checks that every rule is stored this way.

Called by `refresh()`; in js/tests/static_test.js: `badExtras()`, `updaterTests()`
Calls *no function of the code*

5.38 `pictureOf(node, counts)` EXPORTED lines 210–214

Picks an item's icon for data.json from aoe2techtree.net's data: its kind, the picture most civilisations show, and whether it is common, regional or unique, which the outlines and Include Regionals use.

Called by `refresh()`; in js/tests/static_test.js: `badExtras()`
Calls *no function of the code*

5.39 `missingText(set)` EXPORTED lines 216–227

Lists, per language, the page texts and group names that are missing. A test and the updater's report use it, so no language shows a blank.

Called by `compare()`; in js/tests/static_test.js: `textTests()`
Calls `treeKeys()`

5.40 `gather(civs, trees)` INTERNAL lines 229–247

Reads aoe2techtree.net's tree of every civilisation into three tables: each item's node, the civilisations that have it, and how often each of its pictures is used. The updater's first step.

Called by `refresh()`
Calls *no function of the code*

5.41 `addExtras(extras, { nodes, owners, pictures }, say)` INTERNAL lines 249–257

Adds the items of `extras` in data.json to the gathered tables, and reports any that aoe2techtree.net now has itself, so they can be taken out of `extras`.

Called by `refresh()`
Calls `fromExtra()`

5.42 `translate(current, fresh, { keys, nodes, techtree, strings })` INTERNAL lines 259–270

Writes every civilisation's and item's name in every language from the game's texts, unless an override says otherwise, writing out abbreviated names from the help text.

Called by `refresh()`
Calls `capitalise()`, `joinLines()`, `localName()`

5.43 `compare(current, fresh, { keys, nodes, owners, say })` INTERNAL lines 272–291

Lists the changes for the updater's report: civilisations added or gone, items whose civilisations changed or that nobody has, renamed civilisations and items, items not yet in the tree, and missing texts.

Called by `refresh()`
Calls `missingText()`

5.44 `addIcons(fresh, say)` INTERNAL lines 293–301

Adds every new picture to the list of its atlas in `icons.atlas`, in file-name order, and reports it with its PNG, so the atlas can be built again.

Called by `refresh()`
Calls `byFile()`, `iconFile()`

5.45 `refresh(current, techtree, trees, strings)` INTERNAL

lines 303–328

The updater's core: works out the civilisations, who has what, the icons and every name from aoe2techtree.net's data and the extras, dates the data, and writes the report. Stops if the tree names an unknown item, or one twice.

Called by `fetchUpdate()`
Calls `fill()`, `gather()`, `addExtras()`, `treeKeys()`, `availabilityOf()`, `pictureOf()`, `addIcons()`, `translate()`, `compare()`

5.46 `fetchUpdate()` EXPORTED

lines 330–345

Downloads aoe2techtree.net's data, trees and texts, runs `refresh()`, and returns the report and the files that changed, data.json always. So a game update needs only a browser.

Called by in js/view/report.js: `runUpdate()`
Calls `load()`, `refresh()`, `pack()`, `toJson()`

6 model.js

Directory: js/model/ 260 lines 44 functions, 20 exported
Imports: js/model/datahandler.js
Imported by: js/view/stringbuilder.js, js/controller/nodetree.js, js/controller/optionshandler.js, js/tests/run_test.js, js/tests/helpers_test.js, js/tests/behaviour_test.js, js/tests/interface_test.js, js/tests/pictures_test.js, js/tests/audit_test.js

The tree and its rules: which items are shown, how marks pass on, what is greyed out, which civilisations fit, and what Filters shows. It never touches the page.

6.1 `setIncluded(on)` EXPORTED lines 10–12

Turns Include Unique and Universal on or off for the next `rebuild()`.

Called by in js/controller/optionshandler.js: `setInclude()`, `start()`
Calls *no function of the code*

6.2 `setRegionals(on)` EXPORTED lines 14–16

Turns Include Regionals on or off for the next `rebuild()`.

Called by in js/controller/optionshandler.js: `setRegional()`, `start()`
Calls *no function of the code*

6.3 `collect(nodes, parent)` INTERNAL lines 18–32

Turns the shown part of data.json's tree into the items the page uses, each with its key, a path that tells copies apart, its set of civilisations (all, for a group) and its links. Stops with an error for an item without availability.

Called by `rebuild()`
Calls in js/model/datahandler.js: `setOf()`

6.4 `whole(item)` EXPORTED line 34

The set of civilisations that have an item and everything under it, worked out once and kept on the item. A row's count of matching civilisations uses it.

Called by `rebuild()`; in js/tests/helpers_test.js: `fixtures()`
Calls *no function of the code*

6.5 `lift(node, kids)` INTERNAL lines 36–37

Hands the rows of a unique item that is left out to its place, noting the item in their path, so that without the Camel Scout the Camel Rider is a row of the Stable. Lent rows are dropped.

Called by `prune()`
Calls *no function of the code*

6.6 `prune(nodes, every = false, regionals = true)` INTERNAL lines 39–50

Leaves out what Options leaves out: the unique groups and unique items (keeping their rows), regional items without Include Regionals, and items that one, all or all but one civilisation have, unless something under them is shown.

Called by `rebuild()`
Calls `lift()`; in js/model/datahandler.js: `size()`, `setOf()`

6.7 `lent(kid)` INTERNAL line 52

Tells whether a row can be lent to an item that stands in for its own: neither unique nor marked `"own"`.

Called by `common()`
Calls *no function of the code*

6.8 `common(node)` INTERNAL line 54

Copies a row and the rows under it that can be lent, marked as copies, for an item that stands in for it.

Called by `borrowed()`
Calls `lent()`

6.9 `borrowed(nodes, key, line = false)` INTERNAL lines 56–63

Finds the item that a `"replaces"` names and returns the rows it lends; none if it counts as a unit or technology, as its rows are its own upgrades.

Called by `variants()`
Calls `common()`

6.10 `named(link)` INTERNAL line 65

Turns a `"replaces"` or `"excludes"` value, one name, a list or nothing, into a list.

Called by `variants()`, `pair()`, `siblings()`
Calls *no function of the code*

6.11 `variants(nodes)` INTERNAL lines 67–70

Adds under every item that names items it stands in for the rows those items lend, in the order named, so their technologies are written only once in data.json.

Called by `rebuild()`
Calls `named()`, `borrowed()`

6.12 `twin()` INTERNAL lines 72–79

Ties every lent row to its original, so both share one mark: marking Two-Man Saw under the Mule Cart marks it under the Lumber Camp too, with one entry in Filters.

Called by `rebuild()`
Calls *no function of the code*

6.13 `pair(link, list)` INTERNAL lines 81–88

Links items both ways by `"excludes"` or `"replaces"`, as rivals or as alternatives.

Called by `rebuild()`
Calls `named()`

6.14 `side(item, rival)` INTERNAL line 90

An item and everything under it, but its rival and what is under that, so the Stable's side leaves out the Xolotl Warrior inside it.

Called by `oppose()`
Calls *no function of the code*

6.15 `oppose()` INTERNAL lines 92–96

Gives every item the opponents that its must have greys out: everything on its rival's side, and the other way round.

Called by `rebuild()`
Calls `side()`

6.16 `siblings()` INTERNAL lines 98–104

Adds to the alternatives of an item that stands in for others the other items that stand in for the same, so marking the Folwark greys out the Settlement too.

Called by `rebuild()`
Calls `named()`

6.17 `rebuild()` EXPORTED lines 106–118

Builds the items afresh from data.json and the Include options, links copies, rivals and alternatives, and keeps every mark by its path. Runs at the start and whenever an Include option changes.

Called by in js/controller/nodetree.js: `plant()`
Calls `collect()`, `prune()`, `variants()`, `whole()`, `pair()`, `oppose()`, `siblings()`, `twin()`

6.18 `barred(item)` EXPORTED lines 120–121

Tells whether an item is greyed out: not marked must have itself, while one of its alternatives or opponents is (Req. 37).

Called by `locked()`; in js/controller/nodetree.js: `paint()`, `toggle()`
Calls *no function of the code*

6.19 `within(item, top)` INTERNAL line 123

Tells whether an item lies anywhere under another.

Called by `locked()`
Calls *no function of the code*

6.20 `locked(item)` EXPORTED line 125

Tells whether a greyed-out item is also closed: unless its own rival is inside it, as the Xolotl Warrior is inside the Stable, which stays open so the Xolotl Warrior stays in sight.

Called by in js/controller/nodetree.js: `behind()`, `expand()`, `paint()`, `update()`, `wireTools()`
Calls `barred()`, `within()`

6.21 `setMark(item, mark, top = item)` EXPORTED lines 127–130

Marks an item and everything under it, groups only through their rows, skipping the rivals of the item marked. How a building or group is marked or cleared.

Called by `put()`, `unmarkLine()`; in js/controller/nodetree.js: `wireTools()`
Calls *no function of the code*

6.22 `state(item, top = item)` EXPORTED lines 132–136

The mark an item shows: the one its own mark and all its rows share, or `"mixed"` when they differ, leaving out the rivals of the item asked about.

Called by `current()`, `boxState()`, `markOf()`, `filtersOf()`, `chipName()`; in js/tests/behaviour_test.js: `branches()`, `rivals()`
Calls *no function of the code*

6.23 `isUnitOrTech(item)` EXPORTED lines 138–141

Tells whether an item is a unit or technology: an item with another item, or a `"lines"` group, above it. Buildings and groups mark everything in them; units and technologies pass marks along lines.

Called by `prerequisites()`, `current()`, `put()`, `reached()`, `boxState()`, `chipName()`; in js/tests/behaviour_test.js: `standIns()`, `beyond()`, `lines()`
Calls *no function of the code*

6.24 `descendants(item)` INTERNAL line 143

Everything under an item, at every depth.

Called by `put()`, `reached()`, `boxState()`
Calls *no function of the code*

6.25 `covers(above, item)` INTERNAL line 145

Tells whether every civilisation that has an item also has the one above it, so that must have may pass up to it.

Called by `prerequisites()`
Calls *no function of the code*

6.26 `prerequisites(item)` INTERNAL lines 147–151

The units and technologies above an item that its must have passes up to, stopping at a building or at an item that some civilisations with it lack (Req. 34).

Called by `put()`, `mark()`
Calls `isUnitOrTech()`, `covers()`

6.27 `passedTo(item)` INTERNAL lines 153–157

The items a mark passes down to: along the item's line until it forks, and only those the same civilisations have (Req. 29, 30).

Called by `put()`, `reached()`
Calls *no function of the code*

6.28 `current(item)` INTERNAL line 159

The mark an item has now: its own for a unit or technology, otherwise what all of it shows.

Called by `mark()`
Calls `isUnitOrTech()`, `state()`

6.29 `put(item, kind)` INTERNAL lines 161–168

Sets or clears a mark: on everything in a building or group, or along a unit's or technology's line. Must lack also clears must have below that depended on it.

Called by `mark()`
Calls `isUnitOrTech()`, `setMark()`, `descendants()`, `prerequisites()`, `passedTo()`

6.30 `reached(item)` INTERNAL line 170

The rows a mark reached, which marking opens: down to the first fork, and the rows it passed to or everything inside.

Called by `mark()`
Calls `chainOf()`, `isUnitOrTech()`, `passedTo()`, `descendants()`

6.31 `mark(item, kind)` EXPORTED lines 172–177

Marks an item must have or must lack, or clears it if it already has that mark, passes must have up to its prerequisites, and returns the rows to open.

Called by in js/controller/nodetree.js: `toggle()`
Calls `current()`, `put()`, `prerequisites()`, `reached()`

6.32 `boxState(item)` EXPORTED lines 179–182

What a row's box shows: a unit's or technology's own mark, or a dash if a row below has another; otherwise `state()`.

Called by in js/controller/nodetree.js: `paint()`
Calls `isUnitOrTech()`, `state()`, `descendants()`

6.33 `fit()` EXPORTED lines 184–188

Works out the civilisations that fit every mark, as `matching`, and returns the marked items.

Called by in js/controller/nodetree.js: `update()`
Calls *no function of the code*

6.34 `markOf(item)` EXPORTED line 190

The mark an item stands for in Filters: the one all of it shows, else its own.

Called by `filtersOf()`
Calls `state()`

6.35 `lineEnd(item)` EXPORTED lines 192–196

The last item of a line of upgrades that does not fork, or null if the item is a group or its line forks.

Called by `lineOf()`; in js/tests/helpers_test.js: `fixtures()`; in js/tests/behaviour_test.js: `branches()`, `lineTops()`, `counts()`; in js/tests/interface_test.js: `uniqueFilterTests()`
Calls *no function of the code*

6.36 `lineOf(item)` EXPORTED lines 198–205

The whole line of upgrades an item is on, from its top to its end, or just the item if it is on none. Filters names and clears lines with it.

Called by `alone()`, `filtersOf()`, `chipName()`, `unmarkLine()`; in js/tests/helpers_test.js: `lines()`, `fixtures()`; in js/tests/behaviour_test.js: `branches()`, `lineTops()`, `lines()`; in js/tests/interface_test.js: `uniqueFilterTests()`
Calls `lineEnd()`

6.37 `chainOf(item)` EXPORTED lines 207–211

An item and the rows below it for as long as each has exactly one row under it. Marking opens these.

Called by `reached()`; in js/controller/nodetree.js: `expand()`
Calls *no function of the code*

6.38 `unique(item)` INTERNAL line 213

Tells whether data.json counts an item as unique.

Called by `alone()`
Calls *no function of the code*

6.39 `alone(item)` INTERNAL line 215

Tells whether an item is unique in a line that is not all unique, so it keeps its own entry in Filters (Req. 36).

Called by `held()`, `chipName()`
Calls `unique()`, `lineOf()`

6.40 `held(item, mark)` INTERNAL line 217

Tells whether a mark keeps an item's own entry in Filters: must have on an item that `alone()` picks.

Called by `filtersOf()`, `unmarkLine()`
Calls `alone()`

6.41 `filtersOf()` EXPORTED lines 219–235

Works out the entries of Filters: one per marked building, group or line, but none for what a marked row above covers, and the unique items' own entries, in the tree's order.

Called by in js/view/stringbuilder.js: `listFilters()`
Calls `state()`, `held()`, `markOf()`, `lineOf()`

6.42 `share()` EXPORTED lines 237–241

Notes the names that more than one row uses, so that Filters can say which one is meant.

Called by in js/view/stringbuilder.js: `label()`
Calls in js/model/datahandler.js: `nameOf()`

6.43 `chipName(item, kind, own = false)` EXPORTED lines 243–251

Names an entry of Filters: a line after its deepest item with must have, a shared name "in" its parent, and "Full" before a building or group marked in full with more than one row (Req. 24).

Called by in js/view/stringbuilder.js: `listFilters()`; in js/tests/behaviour_test.js: `marking()`, `rivals()`
Calls `isUnitOrTech()`, `lineOf()`, `alone()`, `state()`; in js/model/datahandler.js: `nameOf()`, `fill()`

6.44 `unmarkLine(item, kind, own = false)` EXPORTED lines 253–260

Clears what an entry of Filters stands for: the marks of its kind along its line, or the item and everything in it, keeping the unique items' own entries.

Called by in js/controller/nodetree.js: `onChip()`
Calls `held()`, `lineOf()`, `setMark()`

7 stringbuilder.js

Directory: js/view/ 101 lines 14 functions, 12 exported
Imports: js/model/datahandler.js, js/model/model.js
Imported by: js/script.js, js/view/pictures.js, js/view/scrollbars.js, js/view/report.js, js/controller/nodetree.js, js/controller/optionshandler.js, js/tests/run_test.js, js/tests/helpers_test.js, js/tests/behaviour_test.js, js/tests/interface_test.js, js/tests/pictures_test.js, js/tests/audit_test.js

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The page's text: the fixed texts, the texts of the chosen language, the tree's names, the list of civilisations and Filters, the loading screen, and the small helpers every script that touches the page uses.

7.1 `$(id)` EXPORTED line 6

Short for `document.getElementById`, used wherever the page is touched.

Called by `clone()`, `writeFixed()`, `writeTexts()`, `listCivs()`, `listFilters()`, `showLoadError()`, `showLoading()`; in js/view/pictures.js: `showView()`, `setView()`, `wireTip()`, `startPictures()`; in js/view/report.js: `openReport()`, `report()`, `runUpdate()`; in js/controller/nodetree.js: `plant()`, `find()`, `replant()`, `onChip()`, `wireTools()`, `startTree()`; in js/controller/optionshandler.js: `options()`, `setMenu()`, `buildLanguages()`, `showLanguage()`, `setLanguage()`, `tick()`, `showIcon()`, `onPage()`, `finishLoading()`, `checked()`, `wireOptions()`, `onMenuKey()`, `wireLanguages()`, `start()`; in js/tests/helpers_test.js: `picturesOn()`, `simpleSpot()`, `simpleTicked()`, `civLinks()`, `nameFor()`, `type()`, `pick()`, `reset()`, `beside()`, `listed()`; in js/tests/behaviour_test.js: `chipNames()`, `marking()`, `branches()`, `crowding()`, `parents()`, `rivals()`, `standIns()`, `beyond()`, `lines()`, `counts()`, `tree()`, `finding()`; in js/tests/interface_test.js: `menuTests()`, `languageTests()`, `optionTests()`, `menuAudit()`, `menuKeys()`, `languageAudits()`, `optionKeys()`, `optionAudits()`, `filterFocus()`, `keyboard()`, `findKey()`, `reportTests()`, `uniqueFilterTests()`; in js/tests/pictures_test.js: `checked()`, `cornered()`, `offParchment()`, `wrongSizes()`, `pictureLooks()`, `wrongBoxes()`, `unpictured()`, `findLooks()`, `tipLooks()`, `toolLooks()`, `confirming()`, `simpleSwitch()`, `barLooks()`; in js/tests/audit_test.js: `structureChecks()`, `controlChecks()`
Calls *no function of the code*

7.2 `clone(id)` EXPORTED line 8

Copies the element inside a template in index.html. Repeated markup lives in index.html (Rule 2); the code only fills it in.

Called by `civIcon()`, `listCivs()`, `listFilters()`, `showLoading()`; in js/view/scrollbars.js: `startBars()`; in js/view/report.js: `report()`, `runUpdate()`; in js/controller/nodetree.js: `dress()`, `build()`; in js/controller/optionshandler.js: `buildLanguages()`
Calls `$()`

7.3 `simple()` EXPORTED line 10

Tells whether Simple UI is on, from the `pictures` class on the page, which is what shows the pictures. One source of truth, readable from every script.

Called by `listCivs()`; in js/view/pictures.js: `wireTip()`; in js/view/scrollbars.js: `fit()`; in js/controller/nodetree.js: `dress()`, `wireTools()`; in js/controller/optionshandler.js: `showIcon()`, `finishLoading()`, `setSimple()`, `choices()`; in js/tests/run_test.js: `prepare()`; in js/tests/pictures_test.js: `strayBars()`
Calls *no function of the code*

7.4 `place(icon, { column, row, columns, rows })` EXPORTED lines 12–18

Tells an icon which cell of its atlas to show, as four numbers for CSS: its column and row, and the atlas's columns and rows. Scripts give CSS only numbers.

Called by `civIcon()`; in js/controller/nodetree.js: `dress()`
Calls *no function of the code*

7.5 `compose(node, template, values)` EXPORTED lines 20–25

Writes a text with placeholders into an element, putting elements such as the source link where their placeholder is. Lets a translated sentence contain a link.

Called by `writeFixed()`, `writeTexts()`; in js/controller/nodetree.js: `count()`
Calls *no function of the code*

7.6 `writeFixed()` EXPORTED lines 27–36

Writes the texts that never change: the report window's English words, the source's, Microsoft's and the project's links, and Microsoft's notice around its links. Runs once at the start.

Called by in js/controller/optionshandler.js: `start()`
Calls `$()`, `compose()`

7.7 `writeTexts()` EXPORTED lines 38–46

Writes every text of the chosen language: the page's `lang`, each element marked `data-text`, and the footer's source and date (Req. 5, 11).

Called by in js/controller/optionshandler.js: `setLanguage()`
Calls `compose()`, `$()`

7.8 `label()` EXPORTED lines 48–51

Writes every tree item's name in the current language, and notes the names used more than once.

Called by in js/controller/nodetree.js: `replant()`; in js/controller/optionshandler.js: `setLanguage()`
Calls in js/model/datahandler.js: `nameOf()`; in js/model/model.js: `share()`

7.9 `civIcon(civ)` INTERNAL line 53

Makes a civilisation's icon from its template, showing its cell of the civilisations' atlas. Only used without Simple UI.

Called by `listCivs()`
Calls `place()`, `clone()`; in js/model/datahandler.js: `cellOf()`, `civFile()`

7.10 `listCivs()` EXPORTED lines 55–70

Lists the civilisations that fit the marks, translated and sorted (Req. 4), with icons and links to aoe2techtree.net without Simple UI; writes the count, and the hint when the list is empty.

Called by in js/controller/nodetree.js: `update()`
Calls `$()`, `clone()`, `simple()`, `civIcon()`; in js/model/datahandler.js: `civsOf()`, `civName()`, `plural()`

7.11 `listFilters()` EXPORTED lines 72–86

Makes one Filters button per entry (Req. 22, 23), with its mark and name, and a label telling screen readers what a click clears. Hides Filters when nothing is marked.

Called by in js/controller/nodetree.js: `update()`
Calls `$()`, `clone()`; in js/model/datahandler.js: `fill()`; in js/model/model.js: `filtersOf()`, `chipName()`

7.12 `showLoadError(shown = true)` EXPORTED lines 88–90

Shows, or hides, the message that the data or a language could not be loaded.

Called by in js/script.js: `Start-up`; in js/controller/optionshandler.js: `choose()`
Calls `$()`

7.13 `spin(img)` INTERNAL line 94

Gives the loading picture its address, kept in `data-src` so that it loads only when the loading screen shows it.

Called by `showLoading()`
Calls *no function of the code*

7.14 `showLoading(shown = true)` EXPORTED lines 96–101

Shows or hides the loading screen, with its turning picture, made once, and makes the page behind it unreachable while it shows.

Called by in js/script.js: `Start-up`; in js/controller/optionshandler.js: `finishLoading()`, `setSimple()`
Calls `$()`, `spin()`, `clone()`

8 pictures.js

Directory: js/view/ 61 lines 6 functions, 3 exported
Imports: js/model/datahandler.js, js/view/stringbuilder.js, js/view/scrollbars.js, js/controller/nodetree.js, js/controller/optionshandler.js
Imported by: js/script.js, js/tests/run_test.js

What only the pictures need: Civ view, list or tiles, and the tooltip. It also starts the scroll bars.

8.1 `showView(choice)` INTERNAL lines 9–11

Marks the chosen view among the Civ view radio buttons.

Called by `setView()`, `startPictures()`
Calls in js/view/stringbuilder.js: `$()`

8.2 `setView(choice)` EXPORTED lines 13–18

Shows the civilisations as a list or as tiles, remembers the choice and marks it among the radio buttons. Tiles are pictures, so CSS shows them only without Simple UI.

Called by `startPictures()`; in js/tests/run_test.js: `restore()`
Calls `showView()`; in js/model/datahandler.js: `remember()`; in js/view/stringbuilder.js: `$()`

8.3 `setTips(on)` EXPORTED lines 22–26

Switches the tooltips on or off, remembers the choice, and ticks the Tooltips box.

Called by `startPictures()`; in js/tests/run_test.js: `prepare()`, `restore()`
Calls in js/model/datahandler.js: `remember()`; in js/controller/optionshandler.js: `tick()`

8.4 `tipFor(target)` INTERNAL lines 28–32

Returns the tooltip's text for what the mouse is over: the marking hint over the tree, Include's description over Include, or a civilisation's address over its link.

Called by `wireTip()`
Calls in js/controller/nodetree.js: `itemAt()`

8.5 `wireTip()` INTERNAL lines 34–50

Moves the tooltip with the mouse, into the Options window while it is open, with its corner at the cursor picture's corner and inside the window. It shows only for a mouse, without Simple UI and with Tooltips on.

Called by `startPictures()`
Calls `tipFor()`; in js/view/stringbuilder.js: `$()`, `simple()`

8.6 `startPictures()` EXPORTED lines 52–61

Restores Civ view and Tooltips from the saved choices, adds both to the Options window's choices, starts the tooltip and the scroll bars. Runs once, after `start()`.

Called by in js/script.js: `Start-up`
Calls `setView()`, `setTips()`, `showView()`, `wireTip()`; in js/model/datahandler.js: `recall()`; in js/view/stringbuilder.js: `$()`; in js/view/scrollbars.js: `startBars()`; in js/controller/optionshandler.js: `choices()`

9 scrollbars.js

Directory: js/view/ 113 lines 8 functions, 2 exported
Imports: js/model/datahandler.js, js/view/stringbuilder.js
Imported by: js/view/pictures.js, js/tests/pictures_test.js

The game's scroll bars, which replace the browser's without Simple UI. A bar is placed after each list that scrolls, so it does not scroll with it, and CSS draws it from the numbers `fit()` gives it.

9.1 `fit(bar, box)` EXPORTED lines 14–28

Gives a list's bar its place along the list's right edge, its height and the knob's position, as numbers for CSS, and hides the bar with Simple UI or when the list holds no more than it shows, and the knob when the bar is too short for it.

Called by `fitAll()`, `wire()`
Calls in js/view/stringbuilder.js: `simple()`

9.2 `fitAll()` INTERNAL line 30

Fits every bar again. Runs when a list or what is in it changes size, when anything is added or taken out, when Simple UI changes and when the window does.

Called by `the top-level code`, `startBars()`
Calls `fit()`

9.3 `watch(box)` INTERNAL line 34

Has the size observer watch a list and everything directly in it, so a bar is fitted again when they change size.

Called by `the top-level code`, `startBars()`
Calls *no function of the code*

9.4 `hold(part, step)` INTERNAL lines 41–62

Makes an arrow or the bar's body act on a press, and again every 50 milliseconds after 400 while held, keeping hold of the pointer so the press stays while it leaves.

Called by `wire()`
Calls in js/model/datahandler.js: `attempt()`

9.5 `page(box, knob, y)` INTERNAL lines 64–67

Scrolls a list by a page, less 40 pixels, towards a press above or below the knob, and not at all once the knob reaches the pointer.

Called by `wire()`
Calls *no function of the code*

9.6 `drag(knob, box)` INTERNAL lines 69–86

Lets the knob be dragged: the list scrolls as far as the knob moves, scaled from the body's room to the list's.

Called by `wire()`
Calls in js/model/datahandler.js: `attempt()`

9.7 `wire(bar, box)` INTERNAL lines 88–100

Connects a bar: the arrows scroll 40 pixels, the body pages, the knob drags and the wheel over the bar scrolls the list; the list's scrolling fits the bar again.

Called by `startBars()`
Calls `hold()`, `page()`, `drag()`, `fit()`

9.8 `startBars()` EXPORTED lines 102–113

Puts a bar from its template after every list that scrolls, connects it, and fits the bars again whenever anything about them changes. Runs once, from `startPictures()`.

Called by in js/view/pictures.js: `startPictures()`
Calls `wire()`, `watch()`, `fitAll()`; in js/view/stringbuilder.js: `clone()`

10 report.js

Directory: js/view/ 47 lines 3 functions, 3 exported
Imports: js/model/datahandler.js, js/view/stringbuilder.js
Imported by: js/script.js, js/tests/run_test.js, js/tests/interface_test.js

---

The report window of the updater and the tests.

10.1 `openReport(title, status)` EXPORTED

lines 4–11

Opens the report window, used by the tests and the updater, with a title and a status line, and empties it.

Called by `runUpdate()`; in js/tests/run_test.js: `runTests()`; in js/tests/interface_test.js: `reportTests()`
Calls in js/view/stringbuilder.js: `$()`

10.2 `report(line)` EXPORTED

lines 13–17

Adds a line to the report window.

Called by `runUpdate()`; in js/tests/run_test.js: `runTests()`; in js/tests/interface_test.js: `reportTests()`
Calls in js/view/stringbuilder.js: `clone()`, `$()`

10.3 `runUpdate()` EXPORTED

lines 19–47

Runs the updater: shows its progress, lists what changed, and offers each changed file for download. Opened with `?update`; it changes nothing by itself.

Called by in js/script.js: `Start-up`
Calls `openReport()`, `report()`; in js/model/datahandler.js: `fill()`, `fetchUpdate()`; in js/view/stringbuilder.js: `$()`, `clone()`

11 nodetree.js

Directory: js/controller/ 283 lines 27 functions, 9 exported
Imports: js/model/datahandler.js, js/model/model.js, js/view/stringbuilder.js
Imported by: js/view/pictures.js, js/controller/optionshandler.js, js/tests/behaviour_test.js, js/tests/interface_test.js, js/tests/pictures_test.js

The tree on the left: drawing its rows, opening and closing them, their counts, marking with clicks, touch and keys, Find, and the buttons above the tree.

11.1 `isOpen(item)` EXPORTED line 12

Tells whether an item's branch is open, read from its arrow's `aria-expanded`, which is the one place this is kept.

Called by `shown()`, `plant()`, `count()`, `onKey()`, `onClick()`; in js/tests/behaviour_test.js: `rivals()`, `standIns()`, `lines()`, `counts()`, `tree()`; in js/tests/interface_test.js: `keyboard()`, `headingTests()`
Calls *no function of the code*

11.2 `shown(item)` EXPORTED line 14

Tells whether an item's row can be seen, that is, whether every branch above it is open.

Called by `onKey()`; in js/tests/behaviour_test.js: `branches()`, `rivals()`, `lines()`, `finding()`; in js/tests/interface_test.js: `headingTests()`
Calls `isOpen()`

11.3 `behind(item)` INTERNAL line 16

Tells whether an item lies inside a closed greyed-out item, where Find does not look.

Called by `find()`
Calls in js/model/model.js: `locked()`

11.4 `heading(item)` INTERNAL line 18

Tells whether a group is only a heading inside an item, which has no box and cannot be marked.

Called by `focusOf()`, `itemAt()`, `build()`, `paint()`, `onClick()`
Calls *no function of the code*

11.5 `focusOf(item)` INTERNAL line 20

The part of a row that takes focus: a heading's arrow, or any other row's name.

Called by `onKey()`, `wireTools()`
Calls `heading()`

11.6 `itemAt(target)` EXPORTED lines 22–25

Returns the tree item an element belongs to, but not a heading, so one listener on the tree serves every row, and the tooltip knows when it is over the tree.

Called by `onPress()`, `onRelease()`, `onMenu()`, `onClick()`; in js/view/pictures.js: `tipFor()`
Calls `heading()`

11.7 `dress(item)` EXPORTED lines 27–33

Gives a tree row its icon between its box and name, from its cell of its atlas, with its kind and reach for the outline, or takes it away with Simple UI on (Rules 9, 22).

Called by `plant()`; in js/controller/optionshandler.js: `setSimple()`
Calls in js/model/datahandler.js: `cellOf()`; in js/view/stringbuilder.js: `simple()`, `place()`, `clone()`

11.8 `build(list, ul)` INTERNAL lines 35–68

Makes a row for each item from the template, with its arrow, and ids that tie the arrow and the list to the name for screen readers. A heading loses its box. Calls itself for the items below.

Called by `plant()`
Calls `heading()`; in js/view/stringbuilder.js: `clone()`

11.9 `endPress()` INTERNAL lines 70–76

Cancels a long press in progress and its highlight, when the finger moves or lifts, or the tree is drawn again, and notes when a touch ended.

Called by `plant()`, `onPress()`, `onRelease()`, `startTree()`
Calls *no function of the code*

11.10 `plant()` INTERNAL lines 78–87

Draws the tree from the model again, keeping open branches open, and dresses each row. Runs at the start and through `replant()`.

Called by `replant()`, `startTree()`
Calls `isOpen()`, `endPress()`, `build()`, `dress()`, `setOpen()`; in js/model/model.js: `rebuild()`; in js/view/stringbuilder.js: `$()`

11.11 `count(item)` INTERNAL lines 89–104

Writes an item's count of civilisations that fit the marks: its own number, or "own–all" when it is closed with items under it, and a spoken version for screen readers.

Called by `setOpen()`, `update()`
Calls `isOpen()`; in js/model/datahandler.js: `size()`, `plural()`, `fill()`; in js/view/stringbuilder.js: `compose()`

11.12 `setOpen(item, open)` EXPORTED lines 106–111

Opens or closes an item's branch, and rewrites its count, since a closed item shows two numbers.

Called by `plant()`, `expand()`, `update()`, `toggle()`, `find()`, `onKey()`, `onClick()`, `wireTools()`; in js/tests/behaviour_test.js: `branches()`, `rivals()`, `standIns()`, `beyond()`, `lines()`; in js/tests/interface_test.js: `uniqueFilterTests()`, `headingTests()`; in js/tests/pictures_test.js: `toolLooks()`
Calls `count()`

11.13 `expand(item)` INTERNAL lines 113–115

Opens an item and each only child below it, so opening Militia shows its whole line in one step, unless the item is greyed out and closed. Used by the arrows and the Right key.

Called by `onKey()`, `onClick()`
Calls `setOpen()`; in js/model/model.js: `locked()`, `chainOf()`

11.14 `paint()` INTERNAL lines 117–130

Updates every box: its state for the eye, `aria-checked` for screen readers, the hidden "must lack" words, the bold of a row's own mark, and greyed-out rows and arrows.

Called by `update()`
Calls `heading()`; in js/model/model.js: `boxState()`, `barred()`, `locked()`

11.15 `update()` EXPORTED lines 132–139

Redraws all that depends on the marks: closes greyed-out items, then the boxes, the civilisation list and its count, every item's count and Filters. Runs after every change.

Called by `toggle()`, `replant()`, `onChip()`, `wireTools()`; in js/controller/optionshandler.js: `setLanguage()`, `setSimple()`
Calls `setOpen()`, `paint()`, `count()`; in js/model/model.js: `fit()`, `locked()`; in js/view/stringbuilder.js: `listCivs()`, `listFilters()`

11.16 `toggle(item, kind)` INTERNAL lines 141–145

Handles a mark from a click, a key, a right-click or a long press, unless the item is greyed out: calls `mark()`, opens the way to every row it reached (Req. 28, 31), and redraws.

Called by `onKey()`, `onRelease()`, `onMenu()`, `onClick()`
Calls `setOpen()`, `update()`; in js/model/model.js: `barred()`, `mark()`

11.17 `reveal(node)` INTERNAL lines 147–152

Scrolls the tree's own list just enough to show a row, and nothing around it, so the Find box being typed in never moves (Rule 18).

Called by `find()`
Calls *no function of the code*

11.18 `find()` EXPORTED lines 154–164

Runs Find: highlights the matching items, announces how many there are, opens the branches down to them and shows the first (Req. 20, 21).

Called by `replant()`, `wireTools()`; in js/controller/optionshandler.js: `setLanguage()`
Calls `behind()`, `setOpen()`, `reveal()`; in js/model/datahandler.js: `plain()`, `nameOf()`, `plural()`; in js/view/stringbuilder.js: `$()`

11.19 `replant()` EXPORTED lines 166–171

Builds the tree again and redraws it, its names and its counts, and runs Find again if it holds text. Used when an Include box changes.

Called by in js/controller/optionshandler.js: `setInclude()`, `setRegional()`
Calls `plant()`, `update()`, `find()`; in js/view/stringbuilder.js: `label()`, `$()`

11.20 `onKey(e)` INTERNAL lines 173–195

Handles keys in the tree: Space marks must have and X must lack (Req. 3); Up, Down, Home and End move; Right and Left open, enter, leave and close branches (WCAG 2.1.1).

Called by `startTree()`
Calls `toggle()`, `shown()`, `isOpen()`, `expand()`, `setOpen()`, `focusOf()`

11.21 `onPress(e)` INTERNAL lines 197–208

Starts a right-click, or a touch that becomes a long press after half a second, which marks must lack (Req. 3). The row shows as pressed when the long press is ready.

Called by `startTree()`
Calls `endPress()`, `itemAt()`

11.22 `onRelease(e)` INTERNAL lines 210–217

Ends a press: marks the item must lack if the press was ready and ends on the same item, and swallows the click that follows.

Called by `startTree()`
Calls `endPress()`, `itemAt()`, `toggle()`

11.23 `onMenu(e)` INTERNAL lines 219–225

Stops the browser's own menu over the tree, and marks must lack on the menu key. Right-clicks and long presses are left to `onPress()` and `onRelease()`.

Called by `startTree()`
Calls `itemAt()`, `toggle()`

11.24 `onClick(e)` INTERNAL lines 227–233

Opens or closes a branch from its arrow, or a heading from its name, or marks an item must have (Req. 3), unless the click ends a long press.

Called by `startTree()`
Calls `heading()`, `isOpen()`, `setOpen()`, `expand()`, `itemAt()`, `toggle()`

11.25 `onChip(e)` INTERNAL lines 235–243

Clears the marks behind a Filters button, and moves focus to the next button, or to the heading after the last one (WCAG 2.4.3).

Called by `wireTools()`
Calls `update()`; in js/model/model.js: `unmarkLine()`; in js/view/stringbuilder.js: `$()`

11.26 `wireTools()` INTERNAL lines 245–268

Connects Expand all and Collapse all, which hand focus to each other without Simple UI, Clear (Req. 17 to 19), Find with its clearing cross and Enter, and the Filters buttons. Runs once at the start.

Called by `startTree()`
Calls `setOpen()`, `update()`, `find()`, `focusOf()`, `onChip()`; in js/model/model.js: `locked()`, `setMark()`; in js/view/stringbuilder.js: `$()`, `simple()`

11.27 `startTree()` EXPORTED

lines 270–283

Draws the tree and connects it: keys, presses, moves that cancel a long press, the menu and clicks, then the tools around it. Runs once at the start.

Called by in js/controller/optionshandler.js: `start()`

Calls `plant()`, `onKey()`, `onPress()`, `endPress()`, `onRelease()`, `onMenu()`, `onClick()`, `wireTools()`; in js/view/stringbuilder.js: `$()`

12 optionshandler.js

Directory: js/controller/ 257 lines 35 functions, 1 object, 13 exported
Imports: js/model/datahandler.js, js/model/model.js, js/view/stringbuilder.js, js/controller/nodetree.js
Imported by: js/script.js, js/view/pictures.js, js/tests/run_test.js, js/tests/helpers_test.js, js/tests/behaviour_test.js, js/tests/interface_test.js, js/tests/pictures_test.js, js/tests/audit_test.js

The Options window and the choices in it: the language menu, the two Include boxes and Simple UI, with the screen's pictures, the loading screen and the tab icon. It also starts the page with the visitor's saved choices.

12.1 `options()` EXPORTED — line 10

Lists the language menu's buttons. Used by the menu's keys and by the tests.

Called by `showLanguage()`, `picked()`, `onMenuKey()`; in js/tests/helpers_test.js: `pick()`; in js/tests/interface_test.js: `menuTests()`, `optionTests()`, `menuKeys()`; in js/tests/pictures_test.js: `confirming()`; in js/tests/audit_test.js: `structureChecks()`
Calls in js/view/stringbuilder.js: `$()`

12.2 `setMenu(open, focus)` EXPORTED — lines 12–17

Opens or closes the language menu, hiding the settings while it is open so the window keeps its size (Rule 18), and moves focus if asked.

Called by `wireOptions()`, `onMenuKey()`, `wireLanguages()`; in js/tests/helpers_test.js: `reset()`; in js/tests/pictures_test.js: `barLooks()`
Calls in js/view/stringbuilder.js: `$()`

12.3 `buildLanguages()` INTERNAL — lines 19–28

Makes one button per language, named and tagged in its own language, so screen readers say each name right (WCAG 3.1.2). Runs once at the start.

Called by `start()`
Calls in js/view/stringbuilder.js: `$()`, `clone()`

12.4 `showLanguage(code)` INTERNAL — lines 30–33

Shows a language as chosen in the window: its name on the menu's button and its mark in the list. OK makes it the page's language.

Called by `setLanguage()`, `choices()`, `wireLanguages()`
Calls `options()`; in js/view/stringbuilder.js: `$()`

12.5 `setLanguage(code)` EXPORTED — lines 35–43

Shows the page in a language: its texts, the menu's name and mark, the tree's names, then redraws, and runs Find again if it holds text. Runs at the start and when a language is chosen.

Called by `choose()`, `start()`; in js/tests/run_test.js: `restore()`
Calls `showLanguage()`; in js/model/datahandler.js: `useLanguage()`; in js/view/stringbuilder.js: `writeTexts()`, `label()`, `$()`; in js/controller/nodetree.js: `update()`, `find()`

12.6 `tick(id, on)` EXPORTED — lines 45–48

Shows an Options box as ticked or not, for the eye and for screen readers.

Called by `setInclude()`, `setRegional()`, `setSimple()`, `toggle()`, `discard()`, `start()`; in js/view/pictures.js: `setTips()`
Calls in js/view/stringbuilder.js: `$()`

12.7 `setInclude(on)` EXPORTED — lines 50–55

Switches Include Unique and Universal (Req. 13 to 15): saves the choice, ticks the box and builds the tree again, keeping marks.

Called by `choices()`; in js/tests/run_test.js: `prepare()`, `restore()`; in js/tests/behaviour_test.js: `standIns()`, `beyond()`; in js/tests/interface_test.js: `optionTests()`, `optionKeys()`, `optionAudits()`, `uniqueFilterTests()`, `headingTests()`; in js/tests/pictures_test.js: `unpictured()`, `confirming()`
Calls `tick()`; in js/model/datahandler.js: `remember()`; in js/model/model.js: `setIncluded()`; in js/controller/nodetree.js: `replant()`

12.8 `setRegional(on)` EXPORTED lines 57–62

Switches Include Regionals: saves the choice, ticks the box and builds the tree again, keeping marks.

Called by `choices()`; in js/tests/run_test.js: `prepare()`, `restore()`
Calls `tick()`; in js/model/datahandler.js: `remember()`; in js/model/model.js: `setRegionals()`; in js/controller/nodetree.js: `replant()`

12.9 `showIcon()` INTERNAL lines 64–66

Gives the tab its icon from data.json without Simple UI, and the empty icon `data:,` with it, so Simple UI uses no image at all (Req. 33).

Called by `setSimple()`, `start()`
Calls in js/view/stringbuilder.js: `$()`, `simple()`

12.10 `savedSimple()` EXPORTED line 68

Reads the saved Simple UI choice: off unless it was saved as on, so a first visit shows the pictures.

Called by in js/script.js: `Start-up`
Calls in js/model/datahandler.js: `recall()`

12.11 `screenPictures(width = screen.width, height = screen.height, ratio = devicePixelRatio)` EXPORTED lines 70–75

Works out the pictures for the screen from its size in device pixels: the parchment (`fhd`, `qhd`, `uhd` or `ultrawide`, and `fhd` on a phone) and the size of every other picture, `sd` or `4k`.

Called by `setSimple()`; in js/tests/helpers_test.js: `misjudged()`
Calls *no function of the code*

12.12 `rulesIn(rules)` INTERNAL line 87

Lists every rule of a style sheet, with the rules inside media queries.

Called by `filesOf()`
Calls *no function of the code*

12.13 `linksIn(text, base)` INTERNAL line 89

Lists the picture and font addresses in a rule's text, made whole from the style sheet's own address.

Called by `filesOf()`
Calls *no function of the code*

12.14 `filesOf(sheet, use)` INTERNAL lines 91–92

Lists the addresses in a style sheet's rules whose selectors pass a test.

Called by `pictureFiles()`
Calls `rulesIn()`, `linksIn()`

12.15 `rooted(selector)` INTERNAL lines 94–95

Tells whether a selector already holds for `<html>` or `<body>`, as for the parchment and the cursors, whose pictures show at once.

Called by `setSimple()`
Calls in js/model/datahandler.js: `attempt()`

12.16 `fits(src)` INTERNAL lines 97–101

Tells whether a picture is the one for the screen: the parchment of `data-background`, and the size of `data-assets`. Pictures without a size always fit.

Called by `pictureFiles()`
Calls *no function of the code*

12.17 `pictureFiles(use = () => true)` EXPORTED lines 103–104

Lists the pictures and fonts in the style sheets that the screen uses, so the loading screen can wait for them all.

Called by `finishLoading()`, `setSimple()`
Calls `filesOf()`, `fits()`; in js/model/datahandler.js: `attempt()`

12.18 `copy(src)` INTERNAL line 108

Loads a picture once into a kept image, so that the browser has it before the page shows it.

Called by `finishLoading()`, `setSimple()`
Calls *no function of the code*

12.19 `onPage()` INTERNAL line 110

The addresses of the images already in the page, but for the loading screen's own.

Called by `finishLoading()`
Calls in js/view/stringbuilder.js: `$()`

12.20 `done(img)` INTERNAL line 112

Waits for a kept image to be decoded, or to fail, so that waiting never stops.

Called by `finishLoading()`
Calls *no function of the code*

12.21 `ready(pictures, fonts)` INTERNAL lines 114–115

Tells whether every picture and font has loaded already.

Called by `finishLoading()`
Calls *no function of the code*

12.22 `finishLoading()` EXPORTED lines 117–127

Hides the loading screen once every picture and font has loaded, at once if they have already. Without Simple UI only; a later call takes over from an earlier one.

Called by `setSimple()`; in js/script.js: `Start-up`
Calls `onPage()`, `pictureFiles()`, `copy()`, `ready()`, `done()`; in js/view/stringbuilder.js: `simple()`, `showLoading()`, `$()`

12.23 `setSimple(on)` EXPORTED lines 129–141

Switches Simple UI (Req. 32, 33): saves the choice, ticks the box, picks the screen's pictures the first time, shows the loading screen when leaving Simple UI, adds or removes the `pictures` class, sets the tab icon, and dresses the tree's rows and the list again.

Called by `choices()`; in js/script.js: `Start-up`; in js/tests/run_test.js: `prepare()`, `restore()`; in js/tests/interface_test.js: `reportTests()`; in js/tests/pictures_test.js: `barLooks()`, `pictureTests()`
Calls `tick()`, `screenPictures()`, `pictureFiles()`, `rooted()`, `copy()`, `showIcon()`, `finishLoading()`; in js/model/datahandler.js: `remember()`; in js/view/stringbuilder.js: `simple()`, `showLoading()`; in js/controller/nodetree.js: `dress()`, `update()`

12.24 `picked()` INTERNAL line 143

The language marked in the window's list, which OK makes the page's language.

Called by `choices()`
Calls `options()`

12.25 `choices` OBJECT, EXPORTED lines 145–150

Every setting of the Options window as an entry: how to read its value now, set it, and for the language and Civ view read and show what the window shows. `startPictures()` adds Civ view and Tooltips. Cancel and OK go through it.

Called by `shownOf()`, `apply()`, `discard()`; in js/view/pictures.js: `startPictures()`
Calls `choose()`, `picked()`, `showLanguage()`, `setInclude()`, `setRegional()`, `setSimple()`; in js/view/stringbuilder.js: `simple()`

12.26 `checked(id)` INTERNAL line 152

Tells whether an Options box is ticked.

Called by `shownOf()`, `toggle()`
Calls in js/view/stringbuilder.js: `$()`

12.27 `shownOf(id)` INTERNAL line 154

What the window shows for a setting: its own reading, or its box.

Called by `apply()`
Calls `choices()`, `checked()`

12.28 `toggle(id)` INTERNAL line 156

Ticks or unticks an Options box in the window only; OK applies it.

Called by `wireOptions()`
Calls `tick()`, `checked()`

12.29 `apply()` INTERNAL lines 158–160

Applies every setting whose window value differs from the page's, when OK is pressed.

Called by `wireOptions()`
Calls `choices()`, `shownOf()`

12.30 `discard()` INTERNAL lines 162–164

Shows the page's values again in the window, dropping what was changed, each time the window opens.

Called by `wireOptions()`
Calls `choices()`, `tick()`

12.31 `wireOptions()` INTERNAL lines 166–181

Connects the cogwheel button, the Options window, OK and its boxes, by click and by Space. Runs once at the start.

Called by `start()`
Calls `discard()`, `apply()`, `setMenu()`, `toggle()`; in js/view/stringbuilder.js: `$()`

12.32 `choose(code)` INTERNAL lines 183–197

Chooses a language: loads its file if needed, shows it, saves the choice, and puts it in the address (`?lng=`), so a shared link keeps it. Shows the load error if the file cannot be loaded.

Called by `choices()`
Calls `setLanguage()`; in js/model/datahandler.js: `loadLanguage()`, `remember()`, `attempt()`; in js/view/stringbuilder.js: `showLoadError()`

12.33 `onMenuKey(e)` INTERNAL lines 199–218

Handles the language menu's keys: Down and Up open it at the current language and move through it, Home and End jump, and Escape closes it and returns focus (WCAG 2.1.1).

Called by `wireLanguages()`
Calls `options()`, `setMenu()`; in js/view/stringbuilder.js: `$()`

12.34 `onScrollbar(e)` INTERNAL lines 220–221

Tells whether a click landed on the language list's own scroll bar, which Chrome reports as a click, so that scrolling the list does not close it.

Called by `wireLanguages()`
Calls *no function of the code*

12.35 `wireLanguages()` INTERNAL lines 223–240

Connects the language menu: its button, a click on a language or anywhere else in the list (Req. 11, 12), its keys, and closing when focus or a click goes elsewhere. Runs once at the start.

Called by `start()`
Calls `setMenu()`, `onScrollbar()`, `showLanguage()`, `onMenuKey()`; in js/view/stringbuilder.js: `$()`

12.36 `start()` EXPORTED lines 242–257

Starts the page: restores the Include choices, draws the tree, builds the language menu, writes the fixed texts and the tab icon, connects every control, shows the chosen language, and reveals the page.

Called by in js/script.js: `Start-up`
Calls `tick()`, `buildLanguages()`, `showIcon()`, `wireLanguages()`, `wireOptions()`, `setLanguage()`; in js/model/datahandler.js: `recall()`, `preferredLanguage()`; in js/model/model.js: `setIncluded()`, `setRegionals()`; in js/view/stringbuilder.js: `writeFixed()`, `$()`; in js/controller/nodetree.js: `startTree()`

13 run_test.js

Directory: js/tests/ 107 lines 7 functions, 2 exported
Imports: js/model/datahandler.js, js/model/model.js, js/view/stringbuilder.js, js/view/pictures.js, js/view/report.js, js/controller/optionshandler.js, js/tests/helpers_test.js, js/tests/static_test.js, js/tests/behaviour_test.js, js/tests/interface_test.js, js/tests/pictures_test.js
Imported by: js/script.js (only with `?test`)

Runs the tests and shows the report. Loaded only when the address ends in `?test`.

13.1 `merge(where)` INTERNAL lines 19–28

Combines the results into one line per test, failed if any of its checks failed, with the first details and the window's size. A test may check in several states and languages.

Called by `runTests()`
Calls *no function of the code*

13.2 `reportLine(result, tools)` INTERNAL lines 30–31

Writes a test's report line, such as "Test 15 failed: ...", from its number, name and details.

Called by `runTests()`
Calls in js/model/datahandler.js: `fill()`

13.3 `loadFailed()` EXPORTED lines 33–35

Marks the tests as failed when the page could not load, so CI never waits for a result that cannot come.

Called by in js/script.js: `Start-up`
Calls *no function of the code*

13.4 `prepare()` INTERNAL lines 37–47

Notes the visitor's language, options and saved choices, then sets the tests' starting point: Include unticked, Include Regionals ticked, Simple UI and Tooltips on.

Called by `runTests()`
Calls in js/model/datahandler.js: `recall()`; in js/view/stringbuilder.js: `simple()`; in js/view/pictures.js: `setTips()`; in js/controller/optionshandler.js: `setInclude()`, `setRegional()`, `setSimple()`

13.5 `restore(before)` INTERNAL lines 49–59

Puts back the visitor's language, options, view, address and saved choices after the tests.

Called by `runTests()`
Calls in js/model/datahandler.js: `attempt()`, `remember()`; in js/view/pictures.js: `setView()`, `setTips()`; in js/controller/optionshandler.js: `setInclude()`, `setRegional()`, `setSimple()`, `setLanguage()`; in js/tests/helpers_test.js: `reset()`

13.6 `rounds()` INTERNAL lines 61–82

Runs every test in three rounds, functional, accessibility and maintainability, in the order TESTS.md lists them, picking sample items afresh where the tree was rebuilt.

Called by `runTests()`
Calls in js/model/datahandler.js: `load()`; in js/tests/helpers_test.js: `pick()`, `codes()`, `fixtures()`; in js/tests/static_test.js: `styles()`, `scripts()`, `textTests()`, `tagTests()`, `sourceTests()`, `dataTests()`, `updaterTests()`; in js/tests/behaviour_test.js: `behaviourTests()`; in js/tests/interface_test.js: `menuTests()`, `languageTests()`, `optionTests()`, `uniqueFilterTests()`, `headingTests()`, `keyboardTests()`, `accessibilityTests()`, `reportTests()`; in js/tests/pictures_test.js: `pictureTests()`

13.7 `runTests()` EXPORTED lines 84–107

Runs the tests, numbers them in the order they first ran, checks the 100-test limit (Rule 17), marks the page passed or failed for CI, and shows the report, failures first.

Called by in js/script.js: `Start-up`
Calls `prepare()`, `rounds()`, `restore()`, `merge()`, `reportLine()`; in js/model/datahandler.js: `fill()`; in js/view/report.js: `openReport()`, `report()`; in js/tests/helpers_test.js: `check()`

14 helpers_test.js

Directory: js/tests/ 360 lines 70 functions, 58 exported
Imports: js/model/datahandler.js, js/model/model.js, js/view/stringbuilder.js, js/controller/optionshandler.js
Imported by: js/tests/run_test.js, js/tests/static_test.js, js/tests/behaviour_test.js, js/tests/interface_test.js, js/tests/pictures_test.js, js/tests/audit_test.js

Small tools the test files share: recording results, finding and measuring elements, acting like a visitor, judging pictures and scroll bars, and picking sample items.

14.1 `check(name, ok, detail = "")` EXPORTED line 10

Records one test result. Every test reports this way, and each name is one line in the report.

Called by `expect()`; in js/tests/run_test.js: `runTests()`; in js/tests/static_test.js: `updaterTests()`; in js/tests/behaviour_test.js: `marking()`, `branches()`, `gestures()`, `lineTops()`, `parents()`, `rivals()`, `standIns()`, `lines()`, `counts()`, `tree()`, `finding()`; in js/tests/interface_test.js: `menuTests()`, `languageTests()`, `optionTests()`, `menuKeys()`, `optionKeys()`, `filterFocus()`, `keyboard()`, `findKey()`, `reportTests()`, `uniqueFilterTests()`, `headingTests()`; in js/tests/pictures_test.js: `pictureTests()`; in js/tests/audit_test.js: `audit()`
Calls *no function of the code*

14.2 `expect(name, offenders)` EXPORTED line 12

Records a test that passes when its list of offenders is empty, naming the first few, so a failure points at the exact items.

Called by in js/tests/static_test.js: `sourceTests()`, `textTests()`, `tagTests()`, `treeTests()`, `textDataTests()`, `updaterTests()`; in js/tests/behaviour_test.js: `crowding()`, `counts()`, `tree()`; in js/tests/interface_test.js: `spacingTests()`; in js/tests/pictures_test.js: `pictureLooks()`, `pictureTests()`
Calls `check()`

14.3 `wait(ms)` EXPORTED line 14

Waits a number of milliseconds. Used to hold a long press.

Called by `loaded()`; in js/tests/behaviour_test.js: `gestures()`
Calls *no function of the code*

14.4 `settled(img)` INTERNAL lines 16–17

Waits until an image has loaded or failed.

Called by `loaded()`
Calls *no function of the code*

14.5 `loaded(images)` EXPORTED line 19

Waits until every image has loaded or failed, or ten seconds have passed, so the tests check pictures only once they are there.

Called by `frameShape()`; in js/tests/pictures_test.js: `cornered()`
Calls `settled()`, `wait()`

14.6 `phone()` EXPORTED line 21

Tells whether the window uses the phone layout: narrower than 720 px and at least 640 px tall.

Called by `spans()`
Calls *no function of the code*

14.7 `width(selector)` INTERNAL line 23

Measures the width of the first element a selector finds.

Called by `spans()`
Calls *no function of the code*

14.8 `spans()` EXPORTED lines 25–27

Tells whether the footer is as wide as the panes, or hidden in the phone layout.

Called by in js/tests/interface_test.js: `languageTests()`; in js/tests/pictures_test.js: `pictureLooks()`
Calls `phone()`, `width()`

14.9 `need(ok, problem)` EXPORTED line 29

Returns a problem in a list if a condition fails, and an empty list if it holds, so a test can list exactly what went wrong.

Called by `barProblems()`; in js/tests/pictures_test.js: `cornered()`, `wrongSizes()`, `pictureLooks()`, `unpictured()`, `toolLooks()`, `confirming()`, `simpleSwitch()`, `strayBars()`, `barLooks()`

Calls *no function of the code*

14.10 `same(a, b)` EXPORTED line 31

Tells whether two lists are equal, item by item. Used to compare lists of civilisations.

Called by in js/tests/static_test.js: `updaterTests()`; in js/tests/behaviour_test.js: `marking()`, `branches()`, `lineTops()`, `parents()`, `rivals()`, `standIns()`, `beyond()`, `lines()`; in js/tests/interface_test.js: `menuTests()`, `languageTests()`, `optionTests()`, `optionKeys()`, `filterFocus()`, `uniqueFilterTests()`; in js/tests/pictures_test.js: `toolLooks()`, `confirming()`, `simpleSwitch()`

Calls *no function of the code*

14.11 `codes()` EXPORTED line 33

Lists the language codes from data.json.

Called by in js/tests/run_test.js: `rounds()`; in js/tests/static_test.js: `textTests()`, `tagTests()`, `textDataTests()`, `updaterTests()`; in js/tests/interface_test.js: `englishFirst()`, `menuTests()`, `menuKeys()`, `languageAudits()`, `accessibilityTests()`; in js/tests/pictures_test.js: `toolLooks()`, `confirming()`

Calls *no function of the code*

14.12 `ui()` EXPORTED line 35

Returns the page's texts in the current language.

Called by in js/tests/interface_test.js: `menuTests()`; in js/tests/pictures_test.js: `tipLooks()`, `simpleSwitch()`

Calls *no function of the code*

14.13 `box(el)` EXPORTED line 37

Short for an element's `getBoundingClientRect()`: its place and size on the screen.

Called by `barProblems()`; in js/tests/interface_test.js: `menuTests()`, `spacingTests()`, `reportTests()`; in js/tests/pictures_test.js: `cut()`, `cornered()`, `offParchment()`, `findLooks()`, `tipLooks()`, `toolLooks()`, `confirming()`, `barLooks()`

Calls *no function of the code*

14.14 `keys()` EXPORTED line 39

Lists every item in the full tree.

Called by `regionalKeys()`; in js/tests/static_test.js: `sourceTests()`, `textTests()`, `dataTests()`, `updaterTests()`; in js/tests/interface_test.js: `optionTests()`

Calls in js/model/datahandler.js: `treeKeys()`

14.15 `describe(el)` EXPORTED lines 41–44

Names an element briefly, such as `button#clear` or a tree row's text, for failure messages.

Called by `pictures()`, `picturesOn()`, `barProblems()`; in js/tests/interface_test.js: `spacingTests()`; in js/tests/pictures_test.js: `offParchment()`, `strayCursors()`, `wrongBoxes()`, `toolLooks()`, `confirming()`, `strayBars()`, `ownBars()`; in js/tests/audit_test.js: `focusChecks()`, `structureChecks()`, `controlChecks()`, `colourChecks()`, `layoutChecks()`, `mediaChecks()`

Calls *no function of the code*

14.16 `rendered(el)` EXPORTED lines 46–47

Tells whether an element can be seen: laid out, not hidden, and not text for screen readers only. The accessibility checks look at what is rendered.

Called by `everything()`, `focusables()`, `visibleTexts()`; in js/tests/interface_test.js: `optionTests()`; in js/tests/pictures_test.js: `offParchment()`, `findLooks()`, `toolLooks()`, `confirming()`, `simpleSwitch()`, `strayBars()`; in js/tests/audit_test.js: `structureChecks()`, `controlChecks()`

Calls *no function of the code*

14.17 `everything()` EXPORTED line 49

Lists every element that can be seen on the page.

Called by `pictures()`; in js/tests/interface_test.js: `spacingTests()`; in js/tests/pictures_test.js: `strayCursors()`, `wrongSizes()`; in js/tests/audit_test.js: `audit()`
Calls `rendered()`

14.18 `styles(el)` EXPORTED line 51

Gets an element's styles and those of its `::before`, `::after` and `::marker` parts, so decorations are checked too.

Called by `pictured()`, `drawn()`; in js/tests/pictures_test.js: `cut()`, `wrongSizes()`; in js/tests/audit_test.js: `mediaChecks()`
Calls *no function of the code*

14.19 `pictured(el)` EXPORTED lines 53–55

Tells whether an element shows a picture through CSS: a background, list marker, border image, content or cursor, in itself or its parts.

Called by `pictures()`; in js/tests/audit_test.js: `mediaChecks()`
Calls `styles()`

14.20 `grade()` EXPORTED line 61

The size of pictures the page chose, `sd` or `4k`, from `data-assets`.

Called by `barProblems()`; in js/tests/pictures_test.js: `cut()`, `strayCursors()`, `pictureLooks()`, `wrongBoxes()`, `unpictured()`, `tipLooks()`
Calls *no function of the code*

14.21 `fitted(width = screen.width, height = screen.height, ratio = devicePixelRatio)` EXPORTED lines 63–68

Works out on its own, apart from the page's code, which parchment a screen should get from its size and scale, to judge `screenPictures()`.

Called by `misjudged()`; in js/tests/pictures_test.js: `pictureLooks()`
Calls *no function of the code*

14.22 `misjudged()` EXPORTED lines 75–78

Lists the screens, from phones to ultrawide and from 100% to 300%, for which `screenPictures()` picks another parchment or picture size than `fitted()`.

Called by in js/tests/pictures_test.js: `pictureLooks()`
Calls `fitted()`; in js/controller/optionshandler.js: `screenPictures()`

14.23 `fitting(src, background)` EXPORTED lines 80–84

Tells whether a picture's address suits a parchment: its own parchment, or the `_sd` and `_x1` pictures with fhd and qhd and the `_4k` and `_x2` ones with uhd and ultrawide.

Called by in js/tests/pictures_test.js: `wrongSizes()`
Calls *no function of the code*

14.24 `files(value)` INTERNAL line 86

Reads the addresses out of a CSS value.

Called by `urlsOf()`
Calls *no function of the code*

14.25 `urlsOf(style)` EXPORTED lines 88–89

Lists the picture addresses in an element's style: background, border image, list marker, content and cursor.

Called by `frameShape()`, `sprite()`; in js/tests/pictures_test.js: `wrongSizes()`
Calls `files()`

14.26 `tabIcons()` EXPORTED lines 91–92

Lists the tab icons the page asks for, leaving out the empty icon `data:,`, which is no picture.

Called by `pictures()`; in js/tests/pictures_test.js: `pictureLooks()`; in js/tests/audit_test.js: `mediaChecks()`
Calls *no function of the code*

14.27 `pictures()` EXPORTED lines 94–95

Lists every picture on the page: images, tab icons and everything CSS draws with a picture, the page's own background included. Simple UI must leave it empty (Req. 33).

Called by `picturesOn()`; in js/tests/pictures_test.js: `simpleSwitch()`
Calls `tabIcons()`, `everything()`, `pictured()`, `describe()`

14.28 `picturesOn()` EXPORTED lines 97–98

Tells whether the main pictures show: those of the page, of its body and of the Options button.

Called by in js/tests/pictures_test.js: `simpleSwitch()`
Calls `describe()`, `pictures()`; in js/view/stringbuilder.js: `$()`

14.29 `rgb(colour)` INTERNAL line 100

Reads the numbers of a CSS colour.

Called by `yellow()`, `tone()`
Calls *no function of the code*

14.30 `yellow(colour)` EXPORTED lines 102–105

Tells whether a colour is yellow: red and green high and close, blue clearly lower (Req. 20).

Called by in js/tests/behaviour_test.js: `finding()`
Calls `rgb()`

14.31 `tone(colour)` INTERNAL lines 107–113

Names a colour as white, black, green, red or other, loosely enough for light and dark colours.

Called by `drawn()`
Calls `rgb()`

14.32 `drawn(box)` EXPORTED lines 115–123

Names the mark drawn on a checkbox, such as "green tick", "red cross" or "white dash", from the borders of its `::before` and `::after`, leaving out parts that draw nothing (Req. 14, 23, 26).

Called by in js/tests/behaviour_test.js: `branches()`; in js/tests/pictures_test.js: `wrongBoxes()`
Calls `styles()`, `tone()`

14.33 `simpleSpot()` EXPORTED lines 127–130

Returns where the Simple UI box is, to check that switching Simple UI does not move it (Rule 18).

Called by in js/tests/pictures_test.js: `simpleSwitch()`
Calls in js/view/stringbuilder.js: `$()`

14.34 `simpleTicked(on)` EXPORTED lines 132–133

Tells whether the Simple UI box shows the given state, for the eye and for screen readers.

Called by in js/tests/pictures_test.js: `simpleSwitch()`
Calls in js/view/stringbuilder.js: `$()`

14.35 `civLinks()` EXPORTED line 135

Counts the civilisations in the list that are links.

Called by in js/tests/pictures_test.js: `simpleSwitch()`
Calls in js/view/stringbuilder.js: `$()`

14.36 `focusables()` EXPORTED lines 137–138

Lists every element that can be seen and reached with the keyboard.

Called by in js/tests/audit_test.js: `focusChecks()`, `controlChecks()`, `layoutChecks()`
Calls `rendered()`

14.37 `textOf(node)` INTERNAL lines 140–143

Reads an element's text as a screen reader would, leaving out parts marked `aria-hidden`.

Called by `nameFor()`
Calls *no function of the code*

14.38 `tidy(text)` EXPORTED line 145

Turns runs of spaces into single spaces and trims the ends.

Called by `nameFor()`; in js/tests/interface_test.js: `languageTests()`; in js/tests/audit_test.js: `layoutChecks()`
Calls *no function of the code*

14.39 `nameFor(el)` EXPORTED lines 147–153

Works out an element's accessible name much as a browser does: from `aria-labelledby`, `aria-label`, its label or its text. Lets the tests check that controls are named (WCAG 4.1.2).

Called by in js/tests/interface_test.js: `menuTests()`; in js/tests/audit_test.js: `controlChecks()`
Calls `tidy()`, `textOf()`; in js/view/stringbuilder.js: `$()`

14.40 `squash(text)` EXPORTED line 155

Lower-cases a text and keeps only letters and digits, for loose comparisons.

Called by in js/tests/audit_test.js: `controlChecks()`
Calls *no function of the code*

14.41 `visibleTexts(el)` EXPORTED lines 157–164

Lists the visible pieces of text inside an element. Used to find stray placeholders, and to compare names with what can be seen (WCAG 2.5.3).

Called by in js/tests/interface_test.js: `languageTests()`; in js/tests/audit_test.js: `controlChecks()`
Calls `rendered()`

14.42 `keyDown(target, key)` EXPORTED lines 166–168

Sends a key press to an element, as a visitor's keyboard would.

Called by in js/tests/behaviour_test.js: `marking()`, `branches()`, `lineTops()`, `parents()`, `rivals()`, `standIns()`, `beyond()`, `lines()`, `counts()`; in js/tests/interface_test.js: `menuTests()`, `languageTests()`, `menuKeys()`, `languageAudits()`, `optionKeys()`, `filterFocus()`, `keyboard()`, `findKey()`, `headingTests()`; in js/tests/pictures_test.js: `tipLooks()`, `pictureTests()`
Calls *no function of the code*

14.43 `tap(el)` EXPORTED lines 170–174

Clicks an element with mouse pointer events, as a visitor would.

Called by in js/tests/behaviour_test.js: `marking()`, `branches()`, `crowding()`, `lineTops()`, `parents()`, `rivals()`, `standIns()`, `beyond()`, `lines()`, `counts()`; in js/tests/interface_test.js: `menuTests()`, `languageTests()`, `optionTests()`, `menuAudit()`, `languageAudits()`, `spacingTests()`, `optionKeys()`, `uniqueFilterTests()`; in js/tests/pictures_test.js: `toolLooks()`, `simpleSwitch()`, `barLooks()`, `pictureTests()`
Calls *no function of the code*

14.44 `type(text)` EXPORTED lines 176–179

Types a text into Find, as a visitor would.

Called by `reset()`; in js/tests/behaviour_test.js: `finding()`; in js/tests/interface_test.js: `findKey()`; in js/tests/pictures_test.js: `findLooks()`
Calls in js/view/stringbuilder.js: `$()`

14.45 `pick(code)` EXPORTED lines 181–186

Chooses a language through the Options window, its menu and OK, as a visitor would.

Called by in js/tests/run_test.js: `rounds()`; in js/tests/interface_test.js: `menuTests()`, `languageTests()`, `languageAudits()`; in js/tests/pictures_test.js: `toolLooks()`, `confirming()`
Calls in js/view/stringbuilder.js: `$()`; in js/controller/optionshandler.js: `options()`

14.46 `reset()` EXPORTED lines 188–195

Returns the page to its start: Find empty, no marks, every branch closed, the menu and the Options window closed, nothing focused. Tests start from here.

Called by in js/tests/run_test.js: `restore()`; in js/tests/behaviour_test.js: `marking()`, `branches()`, `crowding()`, `gestures()`, `lineTops()`, `parents()`, `rivals()`, `standIns()`, `lines()`, `counts()`, `tree()`, `finding()`; in js/tests/interface_test.js: `menuTests()`, `languageTests()`, `optionTests()`, `menuAudit()`, `menuKeys()`, `languageAudits()`, `spacingTests()`, `optionKeys()`, `optionAudits()`, `filterFocus()`, `keyboard()`, `findKey()`, `uniqueFilterTests()`, `headingTests()`; in js/tests/pictures_test.js: `barLooks()`, `pictureTests()`
Calls `type()`; in js/view/stringbuilder.js: `$()`; in js/controller/optionshandler.js: `setMenu()`

14.47 `beside()` EXPORTED lines 197–202

Tells whether Find's message is on one line, right of the box and level with it (Req. 21).

Called by in js/tests/behaviour_test.js: `finding()`; in js/tests/pictures_test.js: `findLooks()`
Calls in js/view/stringbuilder.js: `$()`

14.48 `listed()` EXPORTED line 204

Lists the civilisations the page shows.

Called by in js/tests/behaviour_test.js: `marking()`, `crowding()`, `lineTops()`, `parents()`, `rivals()`, `beyond()`, `lines()`; in js/tests/interface_test.js: `menuTests()`, `languageTests()`, `optionTests()`, `optionKeys()`, `filterFocus()`
Calls in js/view/stringbuilder.js: `$()`

14.49 `expected(set)` EXPORTED line 206

Lists the names a set of civilisations should show, translated and sorted, to compare with `listed()`.

Called by in js/tests/behaviour_test.js: `marking()`, `lineTops()`, `parents()`, `rivals()`, `beyond()`, `lines()`; in js/tests/interface_test.js: `menuTests()`, `languageTests()`, `filterFocus()`
Calls in js/model/datahandler.js: `civsOf()`, `civName()`

14.50 `below(item, top)` EXPORTED lines 208–211

Tells whether an item sits anywhere under another.

Called by in js/tests/behaviour_test.js: `branches()`, `parents()`, `standIns()`
Calls *no function of the code*

14.51 `itemAbove(item)` EXPORTED lines 213–216

Tells whether an item has another item above it, or sits in a group marked `"lines"`, which makes it a unit or technology. The tests work this out on their own, apart from the page's code.

Called by `onTop()`, `inner()`, `fixtures()`
Calls *no function of the code*

14.52 `chainOf(item)` EXPORTED lines 218–222

The rows that marking an item should open: the item and those below it while each has one row under it, those with an arrow only. Worked out apart from the page's own `chainOf()`.

Called by in js/tests/behaviour_test.js: `tree()`; in js/tests/interface_test.js: `keyboard()`
Calls *no function of the code*

14.53 `hex(code)` INTERNAL line 226

Turns a colour written as six hex digits into the `rgb()` form the browser reports.

Called by `edges()`
Calls *no function of the code*

14.54 `edges()` EXPORTED lines 228–233

Reads `colour_key.csv` into the outline colour of each kind and reach of tree icon, so the tests compare the page with the file.

Called by in js/tests/pictures_test.js: `unpictured()`
Calls `hex()`; in js/model/datahandler.js: `load()`

14.55 `painted()` EXPORTED line 235

Waits until the browser has drawn the page once more, so that the scroll bars have been fitted.

Called by in js/tests/pictures_test.js: `strayBars()`
Calls *no function of the code*

14.56 `frameShape(el, columns, rows)` INTERNAL lines 237–241

The height of one picture in a sheet divided by its width, from the sheet's own size, so a test can tell a picture drawn in proportion from a stretched one.

Called by `barProblems()`
Calls `urlsOf()`, `loaded()`

14.57 `sprite(el)` INTERNAL lines 243–247

Describes how an element draws its picture: the file, its size, its repeat and its position.

Called by `barProblems()`
Calls `urlsOf()`

14.58 `barProblems(bar, list)` EXPORTED lines 249–278

Lists what is wrong with a list's scroll bar: not along its right edge in the room kept for it, the wrong pictures or the wrong part of them, arrows, body and knob not equally wide, out of proportion, or the knob not where the list is scrolled to.

Called by in js/tests/pictures_test.js: `strayBars()`
Calls `describe()`, `grade()`, `box()`, `frameShape()`, `need()`, `sprite()`

14.59 `shownKeys()` EXPORTED line 280

Lists the items currently in the tree, to compare the default tree with the full one.

Called by in js/tests/interface_test.js: `optionTests()`, `optionKeys()`
Calls *no function of the code*

14.60 `uniqueKeys()` EXPORTED lines 282–289

Lists the items the default tree leaves out as unique: those inside unique groups and those aoe2techtree.net calls unique.

Called by in js/tests/interface_test.js: `optionTests()`, `optionKeys()`
Calls *no function of the code*

14.61 `ownKeys()` EXPORTED lines 291–297

Lists the items marked `"own"` and everything under them, which items standing in for theirs never show.

Called by in js/tests/behaviour_test.js: `standIns()`
Calls *no function of the code*

14.62 `alternativesOf(key)` EXPORTED lines 299–307

Lists the items that marking an item must have should grey out, from data.json alone: the items it names, the items that name it, and those that name what it names (Req. 37).

Called by in js/tests/behaviour_test.js: `standIns()`
Calls *no function of the code*

14.63 `regionalKeys()` EXPORTED line 309

Lists the items aoe2techtree.net calls regional, which Include Regionals leaves out when unticked.

Called by in js/tests/interface_test.js: `optionTests()`, `optionKeys()`
Calls `keys()`

14.64 `outside(nodes, regional)` EXPORTED lines 311–313

Lists the items of a tree outside regional items and all under them: what the tree shows with Include ticked and Include Regionals unticked.

Called by in js/tests/interface_test.js: `optionTests()`
Calls *no function of the code*

14.65 `keptPath()` EXPORTED lines 315–316

Finds the path of a leaf to mark while an Include box changes. Paths survive the tree being built again; item objects do not.

Called by in js/tests/interface_test.js: `optionTests()`, `optionKeys()`
Calls *no function of the code*

14.66 `byPath(path)` EXPORTED line 318

Finds the item with a path in the tree as it is now, after it was built again.

Called by in js/tests/behaviour_test.js: `beyond()`; in js/tests/interface_test.js: `optionTests()`, `optionKeys()`
Calls *no function of the code*

14.67 `onTop(item)` INTERNAL line 320

Tells whether an item is the top of its line: a unit or technology right under a building or group. The tests work this out on their own, apart from the page's code (Req. 27).

Called by `inner()`, `fixtures()`
Calls `itemAbove()`

14.68 `lines()` INTERNAL line 322

Lists the items that start a line.

Called by `fixtures()`
Calls in js/model/model.js: `lineOf()`

14.69 `inner(item)` INTERNAL line 324

Tells whether an item is a unit or technology below the top of its line with one item under it, where a mark passes down only at the same count (Req. 29, 30).

Called by `fixtures()`
Calls `itemAbove()`, `onTop()`

14.70 `fixtures()` EXPORTED lines 326–360

Picks sample items from the tree: leaves some civilisations have, buildings, line tops, even and uneven counts, a line in a group of lines, a building some do without, a lone building and an item with a rival. Found from the data, to survive game updates.

Called by in js/tests/run_test.js: `rounds()`; in js/tests/behaviour_test.js: `standIns()`, `lines()`; in js/tests/interface_test.js: `accessibilityTests()`; in js/tests/pictures_test.js: `pictureTests()`
Calls `itemAbove()`, `lines()`, `onTop()`, `inner()`; in js/model/datahandler.js: `size()`, `nameOf()`; in js/model/model.js: `whole()`, `lineOf()`, `lineEnd()`

15 static_test.js

Directory: js/tests/ 347 lines 21 functions, 7 exported
Imports: js/model/datahandler.js, js/tests/helpers_test.js
Imported by: js/tests/run_test.js

The maintainability and data tests: they read the page's own files as text and check the rules, then check data.json and the updater's writing.

15.1 `scripts()` EXPORTED lines 10–35

Reads every script reachable from `js/script.js`, following its imports, and notes which ones load only through a later import, as the tests do.

Called by in js/tests/run_test.js: `rounds()`
Calls in js/model/datahandler.js: `load()`

15.2 `styles()` EXPORTED lines 37–38

Reads every style sheet that index.html links, with its address.

Called by in js/tests/run_test.js: `rounds()`
Calls in js/model/datahandler.js: `load()`

15.3 `linesOfCode(code)` INTERNAL line 45

Counts the lines that are not blank, as Rules 23 and 25 count a function or a file.

Called by `longFunctions()`, `sizeProblems()`
Calls *no function of the code*

15.4 `regexMayStart(last, word)` INTERNAL lines 47–48

Tells whether a / after a given character or word starts a regular expression rather than a division, so `blanked()` skips the right text.

Called by `blanked()`
Calls *no function of the code*

15.5 `closing(code, at)` INTERNAL lines 50–61

Finds where a string or regular expression that starts at a place ends, minding escapes and, for a regular expression, character sets.

Called by `blanked()`
Calls *no function of the code*

15.6 `blanked(code)` INTERNAL lines 63–105

Blanks out the insides of strings, regular expressions and template texts, keeping the code inside `${}` and every line break, so that only real braces and parentheses are left to count.

Called by `functionsOf()`
Calls `regexMayStart()`, `closing()`

15.7 `opensBody(text, at, paren)` INTERNAL lines 107–113

Tells whether a brace opens a function's body: after =>, or after a parameter list that follows a name which is not `if`, `for`, `while`, `switch`, `catch` or `with`.

Called by `functionsOf()`
Calls *no function of the code*

15.8 `functionsOf(code)` INTERNAL lines 115–131

Finds where every function body in a script starts and ends, by pairing its braces in the blanked text.

Called by `longFunctions()`
Calls `blanked()`, `opensBody()`

15.9 `longFunctions(code, lines)` INTERNAL lines 133–143

Names every function over 60 lines of code, with its line (Rule 23).

Called by `sizeProblems()`
Calls `linesOfCode()`, `functionsOf()`

15.10 `sizeProblems(page, sheets, files)` INTERNAL lines 145–164

Lists every file over 400 lines of code, every function over 60 and every script line over 120 characters (Rules 23 to 25).

Called by `sourceTests()`
Calls `linesOfCode()`, `longFunctions()`

15.11 `unspun(sheet)` INTERNAL line 176

Takes the loading picture's spin out of the style sheets' text, the only animation allowed, before they are searched for others.

Called by `sourceTests()`
Calls *no function of the code*

15.12 `badImages(page, sheets)` INTERNAL lines 178–193

Tells whether any picture breaks the rules: media elements, images outside templates or with an address in them, a tab icon in index.html, a picture in CSS outside a `.pictures` rule or outside `assets/images/`, or a font outside `assets/fonts/` (Rules 9, 22).

Called by `sourceTests()`
Calls *no function of the code*

15.13 `sourceTests(page, sheets, files)` EXPORTED lines 195–244

The rules tests: no comments; no CSS or HTML in the scripts, which hand CSS only numbers; where the files are and what they import, the tests named `_test.js` and loading only when testing, and no two of the files with one name (Rule 29); the size limits; which scripts touch the page or store data; images, fonts, sounds, animations and dependencies; and no game data in the scripts.

Called by in js/tests/run_test.js: `rounds()`
Calls `sizeProblems()`, `badImages()`, `unspun()`; in js/tests/helpers_test.js: `expect()`, `keys()`

15.14 `textTests()` EXPORTED lines 246–254

Checks that every language has all of the page's text, and names every item and civilisation.

Called by in js/tests/run_test.js: `rounds()`
Calls in js/model/datahandler.js: `missingText()`; in js/tests/helpers_test.js: `keys()`, `expect()`, `codes()`

15.15 `tagTests()` EXPORTED lines 256–259

Checks that every language has a name and a valid language tag (WCAG 3.1.1).

Called by in js/tests/run_test.js: `rounds()`
Calls in js/tests/helpers_test.js: `expect()`, `codes()`

15.16 `alike(a, b)` INTERNAL line 263

Tells whether two values are equal, by their JSON.

Called by `badExtras()`
Calls *no function of the code*

15.17 `badExtras(all)` INTERNAL lines 265–277

Lists the entries of `extras` that are wrong: not in the tree, without a name number or a list of known civilisations, or whose availability and icon in data.json do not match what `fromExtra()` makes of them.

Called by `treeTests()`
Calls `alike()`; in js/model/datahandler.js: `fromExtra()`, `availabilityOf()`, `pictureOf()`

15.18 `treeTests(all)` INTERNAL lines 279–302

Checks the tree in data.json: every item once, only known keys, flags only where they belong, `"excludes"` and `"replaces"` naming other known items, and every item with its civilisations and its icon.

Called by `dataTests()`
Calls `badExtras()`; in js/tests/helpers_test.js: `expect()`

15.19 `textDataTests(all)` INTERNAL

lines 304–323

Checks the texts in data.json: counted texts with an "other" form, the same placeholders in every language, overrides of known names, and the source, tool texts and ignore list filled in.

Called by `dataTests()`
Calls in js/tests/helpers_test.js: `codes()`, `expect()`

15.20 `dataTests()` EXPORTED

lines 325–329

Runs the checks of the tree and of the texts in data.json.

Called by in js/tests/run_test.js: `rounds()`
Calls `treeTests()`, `textDataTests()`; in js/tests/helpers_test.js: `keys()`

15.21 `updaterTests()` EXPORTED

lines 331–347

Checks the updater's writing without going online: the files it writes read back the same, availability is stored in its shortest form, and stored names are already tidy.

Called by in js/tests/run_test.js: `rounds()`
Calls in js/model/datahandler.js: `pack()`, `toJson()`, `unpack()`, `addLanguage()`, `localeFile()`, `availabilityOf()`, `civsOf()`, `setOf()`, `joinLines()`; in js/tests/helpers_test.js: `codes()`, `check()`, `same()`, `expect()`, `keys()`

16 behaviour_test.js

Directory: js/tests/ 427 lines 17 functions, 1 exported
Imports: js/model/datahandler.js, js/model/model.js, js/view/stringbuilder.js, js/controller/nodetree.js, js/controller/optionshandler.js, js/tests/helpers_test.js
Imported by: js/tests/run_test.js

The tests of the tree: marking, Filters, the counts, opening and closing, and Find. Each test function names its checks after TESTS.md, which says on which lines each test is found.

16.1 `sameCivs(item)` INTERNAL

lines 11–15

Lists the items of a line, from an item down while each has one item under it that is not a group, that have the same civilisations as the item: those a mark on it passes down to (Req. 29, 30).

Called by `lineTops()`, `lines()`
Calls *no function of the code*

16.2 `chipNames()` INTERNAL

line 17

Lists the names on the Filters buttons.

Called by `branches()`, `lineTops()`, `rivals()`, `standIns()`
Calls in js/view/stringbuilder.js: `$()`

16.3 `fullName(item)` INTERNAL

line 19

Returns "Full" and an item's name, as Filters names a fully marked building (Req. 24).

Called by `branches()`
Calls in js/model/datahandler.js: `fill()`, `nameOf()`

16.4 `marking({ partial, other, third })` INTERNAL

lines 21–38

Tests marking by click, X, the menu key and Space, the civilisation list and its count, and the first Filters button (Tests 3 to 7; Req. 3, 4, 22).

Called by `behaviourTests()`
Calls in js/model/datahandler.js: `plural()`, `size()`, `fill()`; in js/model/model.js: `chipName()`; in js/view/stringbuilder.js: `$()`; in js/tests/helpers_test.js: `reset()`, `tap()`, `check()`, `same()`, `listed()`, `expected()`, `keyDown()`

16.5 `branches({ branch, line, group, headed, lone, walled })` INTERNAL

lines 40–88

Tests buildings, groups and lines: what marking them marks and shows, their Filters buttons and names, "Full" only with something under it and not in a group of lines, and the tick, cross and dash (Tests 8 to 14; Req. 14, 23 to 26, 30).

Called by `behaviourTests()`
Calls `chipNames()`, `fullName()`; in js/model/datahandler.js: `nameOf()`; in js/model/model.js: `state()`, `lineOf()`, `lineEnd()`; in js/view/stringbuilder.js: `$()`; in js/controller/nodetree.js: `setOpen()`, `shown()`; in js/tests/helpers_test.js: `reset()`, `below()`, `tap()`, `check()`, `same()`, `drawn()`, `keyDown()`

16.6 `crowding()` INTERNAL

lines 90–127

Tests that a full tree and the longest civilisation list never make the page scroll, crush a pane or overlap, in whatever window the tests run (Test 15; Rule 12).

Called by `behaviourTests()`
Calls in js/model/datahandler.js: `size()`; in js/view/stringbuilder.js: `$()`; in js/tests/helpers_test.js: `reset()`, `tap()`, `listed()`, `expect()`

16.7 `gestures({ third })` INTERNAL

lines 129–146

Tests that a long press with a finger marks must lack, and that a right-click clears it (Test 16; Req. 3).

Called by `behaviourTests()`
Calls in js/tests/helpers_test.js: `reset()`, `wait()`, `check()`

16.8 `lineTops({ line })` INTERNAL

lines 148–170

Tests that marking the top of a line passes the mark down only at the same count, must have or must lack, that marking it again clears it, and that marking a line's last item marks the line before it with one filter (Tests 17, 18; Req. 29, 30).

Called by `behaviourTests()`

Calls `sameCivs()`, `chipNames()`; in js/model/datahandler.js: `nameOf()`; in js/model/model.js: `lineOf()`, `lineEnd()`; in js/tests/helpers_test.js: `reset()`, `tap()`, `same()`, `listed()`, `expected()`, `keyDown()`, `check()`

16.9 `parents({ branch, loose, without })` INTERNAL

lines 172–186

Tests that must have passes up but never to a building, which then shows a dash, and that an item can be must lack while everything under it is must have (Tests 19, 20; Req. 30).

Called by `behaviourTests()`

Calls in js/view/stringbuilder.js: `$()`; in js/tests/helpers_test.js: `reset()`, `below()`, `tap()`, `check()`, `keyDown()`, `listed()`, `same()`, `expected()`

16.10 `rivals({ rival })` INTERNAL

lines 188–229

Tests an item that excludes another, such as the Xolotl Warrior and the Stable: must have on the one greys out the other and everything in it, the one staying in sight and the other's arrow working; must have on any of those greys out the one; what is greyed out cannot be marked; and must lack greys out nothing (Test 21).

Called by `behaviourTests()`

Calls `chipNames()`; in js/model/model.js: `chipName()`, `state()`; in js/view/stringbuilder.js: `$()`; in js/controller/nodetree.js: `setOpen()`, `shown()`, `isOpen()`; in js/tests/helpers_test.js: `reset()`, `check()`, `tap()`, `same()`, `listed()`, `expected()`, `keyDown()`

16.11 `standIns(sample)` INTERNAL

lines 231–292

Tests an item that replaces others, such as the Pasture and the Farm: it holds everything in their buildings but their unique items and groups, as the same items, so marking one marks both with one filter; must have on one greys out and closes every alternative until it is cleared (Test 22; Req. 37).

Called by `behaviourTests()`

Calls `chipNames()`; in js/model/datahandler.js: `nameOf()`; in js/model/model.js: `isUnitOrTech()`; in js/view/stringbuilder.js: `$()`; in js/controller/nodetree.js: `setOpen()`, `isOpen()`; in js/controller/optionshandler.js: `setInclude()`; in js/tests/helpers_test.js: `reset()`, `check()`, `below()`, `tap()`, `keyDown()`, `same()`, `ownKeys()`, `alternativesOf()`, `fixtures()`

16.12 `beyond()` INTERNAL

lines 294–310

With Include ticked, finds an item that some civilisations have without the unit or technology above it, checks that must have on it and must lack on the one above list just those civilisations, and that unticking Include lifts it up with its mark kept. Part of Test 24.

Called by `lines()`

Calls in js/model/datahandler.js: `size()`; in js/model/model.js: `isUnitOrTech()`; in js/view/stringbuilder.js: `$()`; in js/controller/nodetree.js: `setOpen()`; in js/controller/optionshandler.js: `setInclude()`; in js/tests/helpers_test.js: `tap()`, `keyDown()`, `same()`, `listed()`, `expected()`, `byPath()`

16.13 `lines(sample)` INTERNAL

lines 312–339

Tests that marking a unit or technology opens it to its first fork or its end, and that below the top of a line a mark passes down only while the civilisations stay the same (Tests 23, 24; Req. 28 to 31).

Called by `behaviourTests()`

Calls `sameCivs()`, `beyond()`; in js/model/model.js: `lineOf()`, `isUnitOrTech()`; in js/view/stringbuilder.js: `$()`; in js/controller/nodetree.js: `setOpen()`, `isOpen()`, `shown()`; in js/tests/helpers_test.js: `reset()`, `tap()`, `check()`, `keyDown()`, `same()`, `listed()`, `expected()`, `fixtures()`

16.14 `counts({ partial, other, upgraded })` INTERNAL

lines 341–375

Tests the numbers on the items: closed and open, on lines, and with filters (Tests 25 to 28; Req. 3).

Called by `behaviourTests()`

Calls in js/model/datahandler.js: `fill()`, `size()`, `plural()`, `nameOf()`; in js/model/model.js: `lineEnd()`; in js/view/stringbuilder.js: `$()`; in js/controller/nodetree.js: `isOpen()`; in js/tests/helpers_test.js: `reset()`, `expect()`, `check()`, `tap()`, `keyDown()`

16.15 `tree()` INTERNAL lines 377–395

Tests Expand all, Collapse all and the arrows, and that opening a branch also opens its only children (Tests 29, 30; Req. 18, 19).

Called by `behaviourTests()`
Calls in js/model/datahandler.js: `nameOf()`; in js/view/stringbuilder.js: `$()`; in js/controller/nodetree.js: `isOpen()`; in js/tests/helpers_test.js: `reset()`, `check()`, `expect()`, `chainOf()`

16.16 `finding({ partial })` INTERNAL lines 397–412

Tests that Find highlights its matches in yellow, opens the branches down to them and counts them beside the box, and says when nothing matches (Tests 31, 32; Req. 20, 21).

Called by `behaviourTests()`
Calls in js/model/datahandler.js: `nameOf()`, `plain()`, `plural()`; in js/view/stringbuilder.js: `$()`; in js/controller/nodetree.js: `shown()`; in js/tests/helpers_test.js: `reset()`, `type()`, `check()`, `yellow()`, `beside()`

16.17 `behaviourTests(sample)` EXPORTED lines 414–427

Runs the tests of the tree, in the order TESTS.md lists them.

Called by in js/tests/run_test.js: `rounds()`
Calls `marking()`, `branches()`, `crowding()`, `gestures()`, `lineTops()`, `parents()`, `rivals()`, `standIns()`, `lines()`, `counts()`, `tree()`, `finding()`

17 interface_test.js

Directory: js/tests/ 425 lines 18 functions, 8 exported
Imports: js/model/datahandler.js, js/model/model.js, js/view/stringbuilder.js, js/view/report.js, js/controller/nodetree.js, js/controller/optionshandler.js, js/tests/helpers_test.js, js/tests/audit_test.js
Imported by: js/tests/run_test.js

The tests of the language menu, the languages, the Options window, the keyboard and the report window, and the runs of the accessibility audit.

17.1 `englishFirst()` INTERNAL

lines 16–24

Tells whether a first visit, with no saved choice and a German browser, starts in the language the address names, or else in English (Req. 10). Puts the saved choice back afterwards.

Called by `menuTests()`
Calls in js/model/datahandler.js: `recall()`, `remember()`, `preferredLanguage()`; in js/tests/helpers_test.js: `codes()`

17.2 `menuTests({ partial })` EXPORTED

lines 26–72

Tests the language menu: in Options, every language in its own language, closing on Escape, a click or leaving it, a window that never resizes, the language changed on OK and kept in the address, English first, and marks kept (Tests 33 to 37; Req. 8 to 12).

Called by in js/tests/run_test.js: `rounds()`
Calls `englishFirst()`; in js/view/stringbuilder.js: `$()`; in js/controller/optionshandler.js: `options()`; in js/tests/helpers_test.js: `reset()`, `tap()`, `box()`, `check()`, `codes()`, `keyDown()`, `nameFor()`, `ui()`, `same()`, `listed()`, `expected()`, `pick()`

17.3 `languageTests(code, { partial, other })` EXPORTED

lines 74–100

Tests one language: the page's text, the tree's names, no placeholders, the footer's links and Microsoft's notice, and a translated, sorted civilisation list (Tests 38 to 42; Req. 4, 5, 11).

Called by in js/tests/run_test.js: `rounds()`
Calls in js/model/datahandler.js: `nameOf()`, `fill()`; in js/view/stringbuilder.js: `$()`; in js/tests/helpers_test.js: `reset()`, `pick()`, `check()`, `visibleTexts()`, `tidy()`, `spans()`, `tap()`, `keyDown()`, `same()`, `listed()`, `expected()`

17.4 `optionTests()` EXPORTED

lines 102–159

Tests the two Include boxes: what the tree leaves out by default, ticking and unticking with marks kept, the choices remembered, and Options opening and closing properly (Tests 43 to 46; Req. 7, 13 to 16).

Called by in js/tests/run_test.js: `rounds()`
Calls in js/model/datahandler.js: `size()`, `recall()`, `attempt()`; in js/view/stringbuilder.js: `$()`; in js/controller/optionshandler.js: `options()`, `setInclude()`; in js/tests/helpers_test.js: `reset()`, `keys()`, `uniqueKeys()`, `regionalKeys()`, `check()`, `shownKeys()`, `keptPath()`, `tap()`, `byPath()`, `listed()`, `same()`, `rendered()`, `outside()`

17.5 `menuAudit({ partial })` INTERNAL

lines 161–168

Runs the accessibility audit with a mark and the language menu open. This is the audit's first run, so its checks are listed from here.

Called by `accessibilityTests()`
Calls in js/view/stringbuilder.js: `$()`; in js/tests/helpers_test.js: `reset()`, `tap()`; in js/tests/audit_test.js: `audit()`

17.6 `menuKeys()` INTERNAL

lines 170–185

Tests the language menu's keys: Down opens it at the current language, and Down and End move through it (Test 79; WCAG 2.1.1).

Called by `accessibilityTests()`
Calls in js/view/stringbuilder.js: `$()`; in js/controller/optionshandler.js: `options()`; in js/tests/helpers_test.js: `reset()`, `keyDown()`, `codes()`, `check()`

17.7 `languageAudits(code, { partial, other })` INTERNAL lines 187–199

Runs the accessibility audit in one language: as loaded, with filters, and with everything open. The focus checks run only in the first language.

Called by `accessibilityTests()`
Calls in js/view/stringbuilder.js: `$()`; in js/tests/helpers_test.js: `reset()`, `pick()`, `codes()`, `tap()`, `keyDown()`; in js/tests/audit_test.js: `audit()`

17.8 `spacingTests({ partial })` INTERNAL lines 201–211

Tests that wider text spacing, as WCAG 1.4.12 describes it, makes nothing stick out sideways (Test 80).

Called by `accessibilityTests()`
Calls in js/tests/helpers_test.js: `reset()`, `tap()`, `expect()`, `everything()`, `box()`, `describe()`

17.9 `optionKeys()` INTERNAL lines 213–237

Tests that Space ticks and unticks Include and Include Regionals, and that OK then shows or leaves out those items, keeping marks (Test 81; WCAG 2.1.1).

Called by `accessibilityTests()`
Calls in js/view/stringbuilder.js: `$()`; in js/controller/optionshandler.js: `setInclude()`; in js/tests/helpers_test.js: `reset()`, `uniqueKeys()`, `regionalKeys()`, `keptPath()`, `tap()`, `byPath()`, `listed()`, `keyDown()`, `shownKeys()`, `check()`, `same()`

17.10 `optionAudits()` INTERNAL lines 239–249

Runs the audit with the Options window open, and with everything included and open.

Called by `accessibilityTests()`
Calls in js/view/stringbuilder.js: `$()`; in js/controller/optionshandler.js: `setInclude()`; in js/tests/helpers_test.js: `reset()`; in js/tests/audit_test.js: `audit()`

17.11 `filterFocus({ other, third })` INTERNAL lines 251–267

Tests that a Filters button clears its item and hands focus to the next button, and that clearing the last one hands it to the civilisations' heading, with every civilisation listed (Tests 57, 58).

Called by `keyboardTests()`
Calls in js/view/stringbuilder.js: `$()`; in js/tests/helpers_test.js: `reset()`, `keyDown()`, `check()`, `same()`, `listed()`, `expected()`

17.12 `keyboard({ line })` INTERNAL lines 269–298

Tests the arrow keys in the tree: Right opens a line all the way down, or opens a branch and goes into it, Left goes back out and closes it, Down and Up step between items, End and Home go to the last and the first, and Down on an arrow goes to the next arrow (Tests 59 to 62; WCAG 2.1.1).

Called by `keyboardTests()`
Calls in js/view/stringbuilder.js: `$()`; in js/controller/nodetree.js: `isOpen()`; in js/tests/helpers_test.js: `reset()`, `keyDown()`, `check()`, `chainOf()`

17.13 `findKey({ partial })` INTERNAL lines 300–307

Tests that Enter in Find goes to the first match (Test 63).

Called by `keyboardTests()`
Calls in js/model/datahandler.js: `nameOf()`, `plain()`; in js/view/stringbuilder.js: `$()`; in js/tests/helpers_test.js: `reset()`, `type()`, `keyDown()`, `check()`

17.14 `keyboardTests(sample)` EXPORTED lines 309–313

Runs the keyboard tests: `filterFocus()`, `keyboard()` and `findKey()`.

Called by in js/tests/run_test.js: `rounds()`
Calls `filterFocus()`, `keyboard()`, `findKey()`

17.15 `accessibilityTests(sample)` EXPORTED lines 315–325

Runs this file's audits and accessibility tests in the order TESTS.md lists them: the menu, the first language, spacing and keys, then every other language and the Options window.

Called by in js/tests/run_test.js: `rounds()`
Calls `menuAudit()`, `menuKeys()`, `languageAudits()`, `spacingTests()`, `optionKeys()`, `optionAudits()`; in js/tests/helpers_test.js: `codes()`, `fixtures()`

17.16 `reportTests()` EXPORTED lines 327–354

Tests the report window with 200 lines in it: its title, its count and Close stay in view and only its list scrolls, and Close has focus, sits on the right and is the plain button, with or without Simple UI (Test 99).

Called by in js/tests/run_test.js: `rounds()`
Calls in js/view/stringbuilder.js: `$()`; in js/view/report.js: `openReport()`, `report()`; in js/controller/optionshandler.js: `setSimple()`; in js/tests/helpers_test.js: `box()`, `check()`

17.17 `uniqueFilterTests()` EXPORTED lines 356–394

Tests that a unique item marked must have gets its own Filters button, even in a full group, which clears only it, while its line's or group's button leaves it marked, and that a wholly unique line has one button (Test 54; Req. 36).

Called by in js/tests/run_test.js: `rounds()`
Calls in js/model/datahandler.js: `nameOf()`; in js/model/model.js: `lineOf()`, `lineEnd()`; in js/view/stringbuilder.js: `$()`; in js/controller/nodetree.js: `setOpen()`; in js/controller/optionshandler.js: `setInclude()`; in js/tests/helpers_test.js: `reset()`, `check()`, `tap()`, `same()`

17.18 `headingTests()` EXPORTED lines 396–425

Tests a group inside a building, such as the Castle's Unique Units: it has no checkbox, a click on its name or arrow opens and closes it, a right-click marks nothing, and Down from the row above reaches its arrow (Test 55).

Called by in js/tests/run_test.js: `rounds()`
Calls in js/controller/nodetree.js: `setOpen()`, `isOpen()`, `shown()`; in js/controller/optionshandler.js: `setInclude()`; in js/tests/helpers_test.js: `reset()`, `check()`, `keyDown()`

18 pictures_test.js

Directory: js/tests/ 441 lines 26 functions, 1 exported
Imports: js/model/datahandler.js, js/model/model.js, js/view/stringbuilder.js, js/view/scrollbars.js, js/controller/nodetree.js, js/controller/optionshandler.js, js/tests/helpers_test.js
Imported by: js/tests/run_test.js

The tests of the pictures, run with Simple UI off. There the page need not meet WCAG (Rule 22), nor the UI particulars of REQUIREMENTS.md (Req. 0).

18.1 `at(url, path)` INTERNAL

line 15

Tells whether a CSS `url()` points at a given picture in `assets/images/`.

Called by `picture()`, `findLooks()`
Calls *no function of the code*

18.2 `picture(style, path)` INTERNAL

line 17

Tells whether an element's background is a given picture.

Called by `cut()`, `wrongSizes()`, `pictureLooks()`, `wrongBoxes()`, `offCell()`, `findLooks()`
Calls `at()`

18.3 `cut(el)` INTERNAL

lines 19–22

Tells whether a button is not drawn as the whole button picture: the wrong file or sheet size, drawn parts left on, or not 290 by 50 in proportion.

Called by `toolLooks()`, `confirming()`
Calls `picture()`; in js/tests/helpers_test.js: `grade()`, `styles()`, `box()`

18.4 `shaded(el)` INTERNAL

line 24

Tells whether a button or anything in it has a text shadow, which no button may have.

Called by `confirming()`
Calls *no function of the code*

18.5 `checked(id)` INTERNAL

line 26

Tells whether an Options box with an id is ticked, as it tells screen readers.

Called by `confirming()`
Calls in js/view/stringbuilder.js: `$()`

18.6 `cornered(cancel, ok)` INTERNAL

lines 28–46

Lists what is wrong with where the Options window's buttons are: the panel not between the title and Cancel, or OK not as far from the parchment's right edge as from its bottom edge.

Called by `confirming()`
Calls in js/view/stringbuilder.js: `$()`; in js/tests/helpers_test.js: `loaded()`, `box()`, `need()`

18.7 `offParchment(style)` INTERNAL

lines 48–66

Lists the header, panes and footer that stick out of the parchment, worked out from the five numbers `--ratio`, `--left`, `--top`, `--right` and `--bottom` in pictures.css and the window's size.

Called by `pictureLooks()`
Calls in js/view/stringbuilder.js: `$()`; in js/tests/helpers_test.js: `box()`, `rendered()`, `describe()`

18.8 `strayCursors()` INTERNAL

lines 68–75

Lists the elements without the right cursor: the pointing hand over links and the sword elsewhere, `_x1` at their own size or `_x2` at half (Test 47).

Called by `pictureLooks()`
Calls in js/tests/helpers_test.js: `grade()`, `everything()`, `describe()`

18.9 `wrongSizes()` INTERNAL lines 77–90

Lists the pictures that do not suit the parchment chosen, trying every parchment in turn: `_sd` and `_x1` pictures with fhd and qhd, `_4k` and `_x2` ones with uhd and ultrawide.

Called by `pictureLooks()`
Calls `picture()`; in js/view/stringbuilder.js: `$()`; in js/tests/helpers_test.js: `everything()`, `styles()`, `urlsOf()`, `fitting()`, `need()`

18.10 `pictureLooks()` INTERNAL lines 92–111

Tests the pictures as a whole: the choice for each screen, background, cursors, the Options button and its ring, the tab icon, the parchment and the footer (Test 47).

Called by `pictureTests()`
Calls `picture()`, `wrongSizes()`, `offParchment()`, `strayCursors()`; in js/view/stringbuilder.js: `$()`; in js/tests/helpers_test.js: `fitted()`, `expect()`, `need()`, `misjudged()`, `grade()`, `tabIcons()`, `spans()`

18.11 `wrongBoxes()` INTERNAL lines 113–129

Lists the checkboxes without the sheet's picture for their state, empty, ticked or crossed, or with a drawn mark other than the dash, the radio buttons without theirs, and any state with no box (Test 48).

Called by `pictureTests()`
Calls `picture()`; in js/view/stringbuilder.js: `$()`; in js/tests/helpers_test.js: `grade()`, `drawn()`, `describe()`

18.12 `cellIn(names, name, side)` INTERNAL lines 131–136

Works out where a picture's cell is in its atlas: the files sorted by name, row by row, in a square grid.

Called by `unpictured()`
Calls *no function of the code*

18.13 `share(part, of)` INTERNAL line 138

Turns a column or row into the percentage CSS needs to show that cell of an atlas.

Called by `offCell()`
Calls *no function of the code*

18.14 `offCell(icon, cell, path, grids)` INTERNAL lines 140–148

Tells whether an icon does not show its cell of its atlas: the wrong file, repeat, box, size or position. Notes the atlas for `misfit()`.

Called by `unpictured()`
Calls `picture()`, `share()`

18.15 `misfit(grids)` INTERNAL lines 150–155

Lists the atlases whose size does not fit their grid of 48 px cells, which would shift every icon.

Called by `unpictured()`
Calls *no function of the code*

18.16 `unpictured()` INTERNAL lines 157–182

With every item included, lists the tree rows without the right icon: its cell of its atlas, 44 px, between box and name, and outlined as `colour_key.csv` says (Test 48).

Called by `pictureTests()`
Calls `offCell()`, `cellIn()`, `misfit()`; in js/model/datahandler.js: `civFile()`; in js/view/stringbuilder.js: `$()`; in js/controller/optionshandler.js: `setInclude()`; in js/tests/helpers_test.js: `edges()`, `grade()`, `need()`

18.17 `findLooks()` INTERNAL lines 184–203

Tests the Find box with the pictures: its frame, search icon and clearing cross, its width of up to 325 px, the message beside it, and that matches are marked (Test 49; Req. 21).

Called by `pictureTests()`
Calls `at()`, `picture()`; in js/model/datahandler.js: `nameOf()`; in js/view/stringbuilder.js: `$()`; in js/tests/helpers_test.js: `box()`, `rendered()`, `type()`, `beside()`

18.18 `hover(el, x, y)` INTERNAL lines 205–207

Moves the mouse over an element, as a visitor would.

Called by `tipLooks()`
Calls *no function of the code*

18.19 `tipLooks()` INTERNAL lines 209–233

Tests the tooltip: over the tree at the cursor's corner, over Include with its description, which then shows only there, and none when Tooltips is unticked (Test 49).

Called by `pictureTests()`
Calls `hover()`; in js/model/datahandler.js: `recall()`, `attempt()`; in js/view/stringbuilder.js: `$()`; in js/tests/helpers_test.js: `box()`, `grade()`, `ui()`, `keyDown()`

18.20 `toolLooks()` INTERNAL lines 235–280

Lists what is wrong with the tree's buttons: not Clear and Expand all, or Collapse all while a branch is open, not side by side as wide as the pane, not the whole button picture, text cut, moved, focus not handed over, or different in another language (Test 50).

Called by `pictureTests()`
Calls `cut()`; in js/view/stringbuilder.js: `$()`; in js/controller/nodetree.js: `setOpen()`; in js/tests/helpers_test.js: `box()`, `rendered()`, `need()`, `describe()`, `same()`, `tap()`, `codes()`, `pick()`

18.21 `confirming()` INTERNAL lines 282–324

Lists what is wrong with the Options window's foot and its OK: Cancel and OK not side by side where they belong, button pictures or shadows wrong, or changes taking effect before OK, or kept after Cancel or closing the window (Test 51).

Called by `pictureTests()`
Calls `checked()`, `cornered()`, `cut()`, `shaded()`; in js/view/stringbuilder.js: `$()`; in js/controller/optionshandler.js: `options()`, `setInclude()`; in js/tests/helpers_test.js: `codes()`, `rendered()`, `box()`, `need()`, `same()`, `describe()`, `pick()`

18.22 `simpleSwitch()` INTERNAL lines 326–359

Ticks and unticks Simple UI, and lists what went wrong: a picture, tab icon or link left, the box moved, the choice not remembered, or something not back (Test 52).

Called by `pictureTests()`
Calls in js/model/datahandler.js: `size()`, `recall()`, `attempt()`; in js/view/stringbuilder.js: `$()`; in js/tests/helpers_test.js: `tap()`, `pictures()`, `simpleSpot()`, `picturesOn()`, `civLinks()`, `simpleTicked()`, `same()`, `rendered()`, `need()`, `ui()`

18.23 `strayBars(when)` INTERNAL lines 361–370

Lists the lists whose scroll bar is wrong: one shown where nothing scrolls or with Simple UI, none where the list scrolls, or one that `barProblems()` finds fault with.

Called by `barLooks()`
Calls in js/view/stringbuilder.js: `simple()`; in js/tests/helpers_test.js: `painted()`, `rendered()`, `need()`, `describe()`, `barProblems()`

18.24 `ownBars(width)` INTERNAL lines 372–373

Lists the lists whose own browser scroll bar is not as wanted: hidden with the pictures and back with Simple UI.

Called by `barLooks()`
Calls in js/tests/helpers_test.js: `describe()`

18.25 `barLooks()` INTERNAL lines 375–401

Tests the pictures' scroll bars with every branch closed and open, scrolled to the end, in Options, with the language list and with Simple UI, and that the down arrow and the wheel scroll (Test 53).

Called by `pictureTests()`
Calls `ownBars()`, `strayBars()`; in js/view/stringbuilder.js: `$()`; in js/controller/optionshandler.js: `setMenu()`, `setSimple()`; in js/tests/helpers_test.js: `box()`, `tap()`, `need()`, `reset()`

18.26 `pictureTests()` EXPORTED lines 403–440

Runs the tests of the pictures with Simple UI off, then turns it on again (Tests 47 to 53; Rules 9, 18, 22; Req. 0, 32, 33).

Called by in js/tests/run_test.js: `rounds()`
Calls `pictureLooks()`, `wrongBoxes()`, `unpictured()`, `findLooks()`, `tipLooks()`, `toolLooks()`, `confirming()`, `simpleSwitch()`, `barLooks()`; in js/controller/optionshandler.js: `setSimple()`; in js/tests/helpers_test.js: `reset()`, `fixtures()`, `tap()`, `keyDown()`, `check()`, `expect()`

19 audit_test.js

Directory: js/tests/ 181 lines 14 functions, 1 exported
Imports: js/model/datahandler.js, js/model/model.js, js/view/stringbuilder.js, js/controller/optionshandler.js, js/tests/helpers_test.js
Imported by: js/tests/interface_test.js

The accessibility audit: checks the page as it stands against WCAG 2.2 AAA, wherever a test calls it. Its checks are Tests 64 to 78.

19.1 `rgba(value)` INTERNAL — lines 12–15

Reads a CSS colour into its red, green, blue and alpha numbers.

Called by `backgroundOf()`, `contrastOn()`
Calls *no function of the code*

19.2 `blend([r, g, b, a], [R, G, B])` INTERNAL — line 17

Blends a see-through colour over the colour below it, to get the colour actually seen.

Called by `backgroundOf()`, `contrastOn()`
Calls *no function of the code*

19.3 `backgroundOf(el)` INTERNAL — lines 19–27

Works out the colour behind an element by blending its own and its ancestors' backgrounds. Contrast is measured against this.

Called by `colourChecks()`
Calls `rgba()`, `blend()`

19.4 `luminance(rgb)` INTERNAL — lines 29–35

Computes a colour's relative luminance, as WCAG defines it.

Called by `contrast()`
Calls *no function of the code*

19.5 `contrast(a, b)` INTERNAL — lines 37–40

Computes the contrast ratio of two colours, as WCAG defines it, from 1:1 to 21:1.

Called by `contrastOn()`
Calls `luminance()`

19.6 `contrastOn(colour, behind)` INTERNAL — line 42

Computes the contrast of a colour, which may be see-through, over the colour behind it.

Called by `colourChecks()`
Calls `contrast()`, `blend()`, `rgba()`

19.7 `visibleCentre(el)` INTERNAL — lines 44–57

Finds the middle of the visible part of an element's first line, clipped by every scrolling area around it, to see what is drawn there.

Called by `focusChecks()`
Calls *no function of the code*

19.8 `focusChecks(list)` INTERNAL — lines 59–74

Focuses every control in turn, and checks that it shows a thick outline with 3:1 contrast and that nothing covers it (Test 78; WCAG 2.4.7, 2.4.12, 2.4.13).

Called by `audit()`
Calls `visibleCentre()`; in js/tests/helpers_test.js: `focusables()`, `describe()`

19.9 `structureChecks(note, list)` INTERNAL — lines 76–99

Checks the page's title and languages, its landmarks and headings, and that ids are unique and ARIA references resolve (Tests 64 to 68; WCAG 1.3.1, 2.4.2, 3.1.1, 3.1.2, 4.1.2).

Called by `audit()`
Calls in js/view/stringbuilder.js: `$()`; in js/controller/optionshandler.js: `options()`; in js/tests/helpers_test.js: `describe()`, `rendered()`

19.10 `controlChecks(list)` INTERNAL lines 101–116

Checks that boxes, branches and the language menu say their state, and that every control has a name containing its visible text (Tests 69, 70; WCAG 2.5.3, 4.1.2).

Called by `audit()`
Calls in js/model/datahandler.js: `nameOf()`; in js/view/stringbuilder.js: `$()`; in js/tests/helpers_test.js: `describe()`, `focusables()`, `rendered()`, `nameFor()`, `visibleTexts()`, `squash()`

19.11 `colourChecks(all, list)` INTERNAL lines 118–134

Checks the contrast of all visible text, 7:1, or 4.5:1 when large, and of control borders, 3:1 (Tests 71, 72; WCAG 1.4.6, 1.4.11).

Called by `audit()`
Calls `contrastOn()`, `backgroundOf()`; in js/tests/helpers_test.js: `describe()`

19.12 `layoutChecks(all, note, list)` INTERNAL lines 136–155

Checks 44 by 44 targets, no page scrolling, nothing sticking out sideways, and unjustified text with 1.5 line spacing (Tests 73 to 76; WCAG 2.5.5, 1.4.10, 1.4.8; Rule 12).

Called by `audit()`
Calls in js/tests/helpers_test.js: `tidy()`, `focusables()`, `describe()`

19.13 `mediaChecks(all, list)` INTERNAL lines 157–167

Checks that the page shows no media, tab icon or CSS picture, and runs no animations or transitions (Test 77; Rules 9, 10, 11, 22).

Called by `audit()`
Calls in js/tests/helpers_test.js: `tabIcons()`, `pictured()`, `describe()`, `styles()`

19.14 `audit(label, { focus = true } = {})` EXPORTED lines 169–181

Checks the page as it stands through the checks above, and the focus checks if asked, under a name that says the state. Called in many states and languages.

Called by in js/tests/interface_test.js: `menuAudit()`, `languageAudits()`, `optionAudits()`
Calls `structureChecks()`, `controlChecks()`, `colourChecks()`, `layoutChecks()`, `mediaChecks()`, `focusChecks()`; in js/tests/helpers_test.js: `check()`, `everything()`

20 data.json

Directory: the main folder 2251 lines 12 keys

All of the page's data, in English: the civilisations, the tree, who has each item, its icon and the page's texts. `useData()` reads it at start-up, and the updater writes it through `pack()` and `toJson()`: short objects and lists on one line, the rest indented by two spaces. The other languages' texts are in `locales/`, one file each, laid out like the `overrides` and `text` here.

An item is named by its English name throughout: the tree, `availability`, `pictures`, `extras` and `text.items` use the same names as keys.

Lines	Key	What is there
2–11	`source`	Where the data comes from: aoe2techtree.net and its data address, `helpOffset` (taken from an item's help string number to find its text, in `localName()`), the month of the last update, and Microsoft's notice with the rules and the project it links.
12–30	`languages`	Every language by its code, with its name in itself, its language tag, and whether its words are spaced and its hyphens at line ends are merged. English comes first.
31	`civs`	The civilisations, in the order whose bits the sets use. A set is a number with one bit per civilisation (`bit()`).
32–47	`icons`	Where the pictures are: the tab icon, the file names of civilisations that differ from their names, and the folder of the tree's icons.
36–40	`groups`	The icon of each top-level group, by kind and picture.
41–46	`atlas`	For each kind, the pictures in its atlas, which `cellIn()` places by file name.
48–687	`tree`	The tree as nested objects: `{"group": …}` for a top-level group or a group in a building, `{"item": …}` for an item, with `kids` for what is under it and the flags `own`, `lines`, `excludes` and `replaces` where they belong. `rebuild()` reads it.
688–1159	`availability`	Who has each item: `"all"`, `{"has": […]}` or `{"lacks": […]}`, whichever is shortest (`availabilityOf()`). `setOf()` turns it into a set.
1160–1631	`pictures`	Each item's icon: its kind, its picture in that kind's atlas, and its reach, common, regional or unique, which Include Regionals and the outlines use.
1632	`ignore`	Names on aoe2techtree.net that the updater does not report as new items.
1633–1642	`extras`	Items aoe2techtree.net does not show in its tree, with their kind, picture, reach, name number and civilisations. `fromExtra()` turns each into availability and a picture.
1643	`overrides`	Names to use instead of the game's own, where it gets one wrong.
1644–1670	`tools`	The texts of the tests' report and of the updater, in English only, with `{}` placeholders that `fill()` fills in.
1671–2250	`text`	The page's texts in English, in four parts.
1672–1712	`ui`	The page's own texts, by the name `data-text` gives in index.html, counted ones as `{"one": …, "other": …}` for `plural()`.
1713–1719	`groups`	The name of each group.
1720–1777	`civs`	The name of each civilisation.
1778–2249	`items`	The name of each item.

21 de.json

Directory: locales/ 587 lines 2 keys

The German texts. Every other language has a file like it in `locales/`, which the page loads when that language is chosen (`localeFile()`) and the updater writes. It holds the two keys of data.json that differ by language, with the same keys inside.

Lines	Key	What is there
2–6	`overrides`	German names to use instead of the game's own.
7–586	`text`	The page's texts in German, in the same four parts as data.json's `text`.
8–48	`ui`	The page's own texts.
49–55	`groups`	The name of each group.
56–113	`civs`	The name of each civilisation.
114–585	`items`	The name of each item.

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